Deep Learning-Based Biosignal Control of a Soft Hand Exoskeleton for Grasp Assistance

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Abstract
Hand function and independent dexterity are frequently compromised by age-related decline, traumatic injury, and prevalent inflammatory or neurological conditions such as rheumatoid arthritis and carpal tunnel syndrome. This thesis presents the design, fabrication, and testing of SoFiE (Soft Finger Exoskeleton for Intelligent Grasping), a lightweight, modular, 3D-printed soft wearable robotic platform customized for pinch-grasp assistance. Fabricated using variable-shore TPU to optimize structural compliance and user comfort. SoFiE utilizes a single-actuator, underactuated tendon mechanism with a floating pulley to distribute force between the thumb and index finger. Two embedded sensing modalities provide feedback: StretchSense, a conductive flexible spring providing passive extension forces and proxy for position tracking, and MagSense, a fingertip magnet-magnetometer pair that measures contact detection and material compliance. To enable intuitive control, an intention decoding control pipeline was developed using sequential deep learning architectures, including LSTM, CNN+LSTM, and CNN+LSTM with attention, evaluated across EEG and EMG signals. System evaluations across 17 participants proved that while decision-level multi-sensor fusion suffers from low EEG signal-to-noise ratios, an EMG-driven CNN+LSTM network architecture effectively overcomes muscle crosstalk, achieving an exceptional subject-independent classification accuracy ceiling of 99.4%. Real-time online validation coupled with a digital twin yielded an optimal response latency of 0.45 ± 0.79 s. Finally, functional human-in-the-loop grasping evaluations demonstrated that while a passive configuration introduces minimal baseline resistance, active assistance dynamically attenuates user metabolic demand, achieving a statistically profound 92.6% reduction in muscular effort during heavy, object-loaded lifting tasks. These results demonstrate the potential of SoFiE for intuitive, low-effort pinch assistance using real-time biosignal-driven control.

Contents

1. Introduction

The human hand is among the most dexterous structures of the human body, allowing humans to interact with and adapt to the surrounding environment. Through a combination of fine motor control, powerful grasping, and rich tactile feedback, the hand can perform a wide range of daily tasks, from delicate manipulation of small objects to stable handling of larger or heavier objects (Maw et al 2016). In addition to being mechanically important for grasping and manipulation, the hand is central to tactile perception and adaptive interaction with the environment (Fischer et al 2014). Despite this versatility, hand function is vulnerable to impairment caused by neurological disorders, physical injury, repetitive strain, inflammatory disease, and age-related decline (Gopal et al 2022).
Carpal tunnel syndrome (CTS) is a common hand impairment associated with repetitive and monotonous wrist activities, as well as rheumatoid inflammation. CTS can cause pain, numbness, and tingling in the hand, especially in the thumb and index finger, and may reduce grip strength and hand function (Genova et al 2020).
Rheumatoid arthritis (RA), the most common form of inflammatory arthritis, is characterized by immune-mediated damage to joints, muscles, tendons, and bone structures. This leads to inflammation, swelling, pain, and in severe cases permanent disability. RA commonly affects small joints, including those in the hands and feet (Mohammed et al 2020).

CTS, RA, neurological disorders, physical injuries, and age-related weakness can all reduce hand mobility, grip strength, and the ability to perform activities of daily living. These limitations are especially problematic when they affect the pinch grasp, since pinch motion is essential for fine and precise manipulation. Modern grasp taxonomies identify thumb-finger and lateral pinch patterns as important grasp types used in everyday manipulation tasks (Feix et al 2016; Napier 1956) and the coordination between the thumb and index finger allows humans to pick up small objects, button clothes, hold a pen, operate tools, and interact with electronic devices (Duruoz 2014; Kivell 2021).
For people with reduced hand function, loss of pinch strength or pinch coordination can therefore reduce functional independence and quality of life (Hyun et al 2015).
To address hand impairments like RA, clinical care often combines exercise and rest with assistive devices to preserve mobility and minimize joint stress Boer et al 2009. This motivates the development of soft wearable hand exoskeletons capable of assisting grasping and reducing muscular effort. For effective pinch assistance, such devices must provide sufficient force, coordinated thumb-index motion, reliable sensing, and compliant, comfortable hand interaction. Assistive hand devices can be passive or active (Birch et al 2008), depending on whether they support the hand mechanically or provide powered assistance or resistance. One example of an active assistive device is the I-Phlex finger exoskeleton (Figure 1**A**), which demonstrated how active mobilization can increase the range of motion of the knuckles and thereby reduce joint stiffness (Peperoni et al 2024).
More generally, hand exoskeletons have evolved from rigid linkage-based systems toward softer and more wearable architectures. Rigid exoskeletons can provide accurate motion guidance and high force transmission, and they have shown strong potential for rehabilitation and force augmentation (Plessis et al 2021; Sarac et al 2019). However, they often require careful joint alignment, include bulky mechanical structures, and may restrict natural hand motion or reduce comfort during prolonged use.
Soft and compliant hand exoskeletons address several of these limitations by using flexible materials, tendon-driven mechanisms, fabric structures, or fluidic actuators (Figure 1**B**) that better conform to the hand and allow more natural movement (Chu and Patterson 2018; Polygerinos et al 2015). Rudd et al. proposed a sub $200 3D-printed soft hand exoskeleton (Figure 1**C**) for use in hand therapy, providing a simpler, cheaper, and more flexible alternative to many commercially available products (Rudd et al 2019). A review by Heo et al. compares different types of hand exoskeleton technologies for rehabilitation and assistance. Although more than a decade has passed since the publication of the review, many of the principles and mechanisms are still used in current designs (Heo et al 2012). The

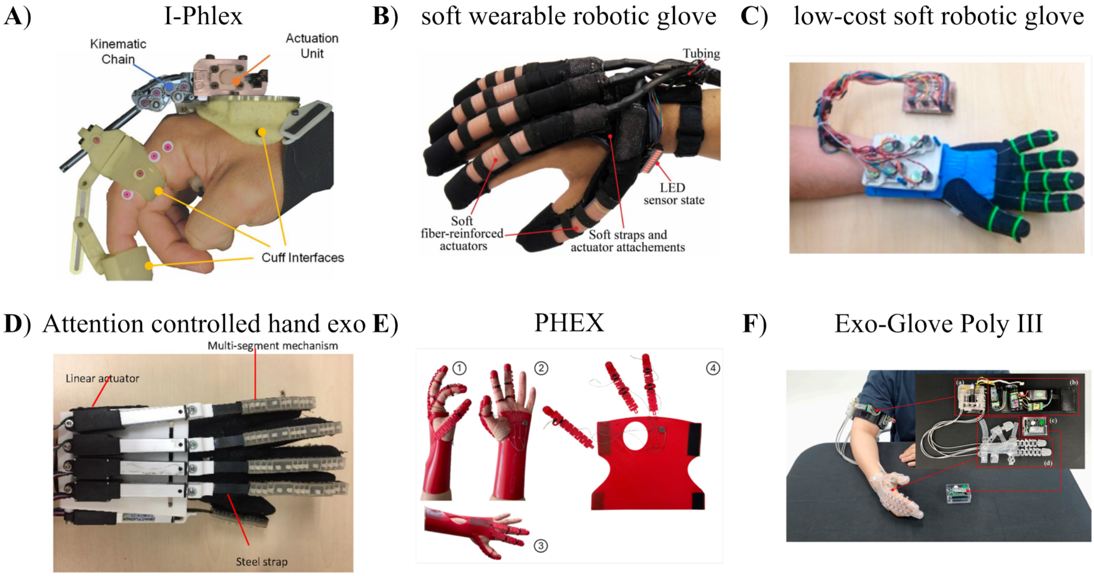


Figure 1. *six examples of wearable robotic gloves.* ***A*** *I-Phlex designed by Peperoni et al. utilizing rigid links to transmit actuation force.* ***B*** *a soft wearable robotic glove designed by Polygerinos et al. using molded elastomeric chambers with fluid pressurization for actuation.* ***C*** *a low-cost soft robotic glove design by Rudd et al. utilizing 3D printing to minimize cost and tendons for actuation.* ***D*** *an attention controlled hybrid rigid-soft hand exoskeleton designed by Li et al. using EEG as a brain-controlled switch to actuate the exoskeleton.* ***E*** *PHEX, a pediatric hand exoskeleton for activities of daily living designed by D'Angelo et al. using a passive cable and spring setup for assisting finger motion.* ***F*** *Exo-Glove Poly III, a three finger exoskeleton designed by Kim et al. utilizing a soft elastomer glove, with a slack-based tendon actuation unit for grasp assistance.*

review highlights that rigid systems are often mechanically complex or physically bulky, which can complicate the development of multi-fingered variants. In contrast, soft and compliant mechanisms can reduce mechanical complexity and improve wearability when combined with careful mechanical design.

Hybrid rigid–soft designs (Figure 1**D**) have also been explored to combine mechanical support with improved wearability and user intention control (Li et al 2019). D'Angelo et al. proposed PHEX (Figure 1**E**), a customized pediatric hand exoskeleton that uses a ribbon kirigami-inspired structure to create a compliant dorsal mechanism. The design also incorporates multi-material cable guide rings made from TPU and Nylon to improve wearer comfort while maintaining structural rigidity (D'Angleo et al 2025). Kim et al. developed the Exo-Glove Poly III (Figure 1**F**), a three-finger soft hand exoskeleton based on an underactuated tendon-driven mechanism. Their design achieves multiple degrees of freedom (DOF) using a single actuator and enables gesture control and manipulation with the thumb, index finger, and middle finger through a slack-based sequential actuation strategy (KB Kim et al 2025). Despite these advances, soft hand exoskeletons still face challenges in compact actuation, reliable force transmission, sensing integration, portability, and intuitive control (Chu and Patterson 2018; Plessis et al 2021). These challenges are especially important for pinch assistance, where the device must remain lightweight and comfortable while coordinating thumb and index finger motion during contact-rich object manipulation (Sarac et al 2019). Many soft systems rely on pneumatic or hydraulic hardware, off-board actuation, or bulky control units, which can limit portability and everyday use.

Other systems improve portability but may still be difficult to customize for individual hand anatomy or require complex fabrication and assembly.

In addition to actuation mechanisms, sensing is an important part of hand exoskeleton design. Reliable information about finger motion, tendon loading, fingertip contact, grasp force, and object compliance is essential for safe and adaptive assistance. Heo et al. discussed several sensing technologies that have been used, or could be used, in hand exoskeletons, including finger pad deformation sensing for estimating gripping force. Nakatani et al. designed a compact sensor placed on top of the fingernail, with T-shaped prongs extending to both sides of the finger pad. These prongs used strain gauges to measure horizontal deformation of the fingertip, while avoiding obstruction of the wearer's natural tactile interaction with the object being touched (Nakatani et al 2011). More recently, Pattabiraman et al. proposed a magnetic touch sensor based on cut-cell microstructures for sensing gripping force and slip in robotic actuators (Pattabiraman et al 2025). While these sensing approaches provide information about the interaction between the hand, exoskeleton, and object, assistive control also requires information about the user's hand movement.

To capture hand movement, researchers frequently turn to physiological biosignals. Among these, electroencephalography (EEG) serves as a method for decoding neural commands directly from the brain. The foundational principles of this technique date back to 1924, when Hans Berger first, recorded the electrical activity of the brain and identified the alpha and beta rhythms as fundamental patterns of brain activity (Tudor et al 2005). The field later developed into brain-computer interfaces (BCIs), which establish a communication pathway between brain activity and external devices without relying on peripheral nerves or muscles (Vidal 1973). BCIs have since been applied in areas such as neurorehabilitation, prosthetic control, emotion recognition, and gaming. However, practical BCI systems remain limited by challenges such as poor portability, low signal-to-noise ratio (SNR), limited spatial resolution, artifacts, inter-session and inter-subject variability, and the trade-off between model complexity and real-time performance (Sarhan et al 2023; Värbu et al 2022).
Additionally, electromyography (EMG) provides a complementary biosignal by measuring the electrical activity generated during muscle contractions. Surface EMG is non-invasive and well suited for motion-related decoding, especially when the goal is to identify active muscles, movement onset, or performed movements. EMG-based pattern recognition has been used for real-time control of hand exoskeletons and has demonstrated potential for supporting grip assistance in users with impaired hand function (Lu et al 2017). However, standalone EMG also has limitations, since signal quality can be affected by electrode placement, crosstalk, motion artifacts, electrical noise, muscle fatigue, paralyzed limbs, neurodegenerative conditions, and inter-subject variability (Karthick and Ramakrishnan 2016; Sarhan et al 2023). These limitations can degrade classification accuracy over time and make generalizable control difficult, especially in real-time wearable applications.

Recent BCI research has therefore shown increasing interest in hybrid systems that combine multiple physiological signals to improve robustness. Pfurtscheller et al. demonstrated a simultaneous hybrid BCI architecture in which brain activity was processed together with another physiological signal. Their hybrid model achieved an accuracy of 84.5 ± 10.2, compared to 69.4 ± 8.6 for the conventional BCI model. Although the improvement in mean accuracy was not statistically significant, the number of participants affected by BCI illiteracy was reduced from 11 to 1, suggesting that hybrid systems can improve robustness across users (Pfurtscheller et al 2010). Combining EEG and EMG is particularly relevant because the two modalities provide different but complementary information. EEG can capture central cues related to motor intention and planning, while EMG reflects peripheral muscle activation during movement execution. In this way, EEG–EMG fusion can potentially connect the user's intended

movement with the corresponding muscular response, thereby improving the reliability of intent detection compared to either modality alone (Sarhan et al 2023).
The temporal nature of EEG and EMG signals makes sequential modeling important for biosignal-based control. These signals vary over time and contain information related to movement onset, sustained activation, and relaxation. Therefore, models used for real-time biosignal classification must be able to extract meaningful temporal features while remaining computationally efficient enough for real-time deployment. Recurrent neural networks, such as long short-term memory (LSTM), have shown strong potential for modeling sequential data and capturing long-term dependencies (Graves 2013; Hochreiter and Schmidhuber 1997). Transformer architectures later showed that self-attention could replace recurrent processing to improve sequence modeling and reduce computational cost in translation tasks (Vaswani et al 2017). For high-frequency data streams, WaveNet demonstrated that dilated causal convolutions can expand the receptive field and process long temporal sequences efficiently (Van Den Oord et al 2016). Within biorobotics, EMG-based gesture recognition has shown that CNNs can effectively decode EMG signals, while recurrent layers remain useful for capturing more complex temporal dynamics (Cornelis et al 2018). However, many deep learning models remain computationally demanding for long sequences and high-dimensional representations, leading to inference bottlenecks that can limit real-time robotic control. This creates a trade-off between model complexity, classification performance, and inference latency, which remains one of the central challenges in practical BCI and hybrid BCI systems (Sarhan et al 2023; Värbu et al 2022).

This thesis presents the development of SoFiE, a soft wearable hand exoskeleton for pinch-grasp assistance, together with a biosignal-based control framework for decoding user movement. The system is designed to assist flexion of the index finger and thumb while remaining lightweight, modular, and suitable for untethered operation. SoFiE will be developed as the physical assistance platform, and primarily fabricated using flexible 3D-printed components. It will be actuated using a tendon-driven mechanism, providing the tension necessary for finger flexion, while sensing will be integrated directly into the mechanical structure of the device with two novel sensors.
StretchSense, a compliant conductive spring element for proprioceptive sensing, and MagSense, a finger tip mounted magnet–magnetometer sensor for contact and deformation related tactile feedback.
The biosignal-based control framework for decoding user movements will be developed with emphasis on low-latency operation and subject-independent performance. Motion recognition will be evaluated using sequential deep learning architectures, including LSTM and CNN+LSTM models, while attention is explored as an additional mechanism. The functional effect of the exoskeleton will be assessed by comparing muscular effort across three conditions: no exoskeleton, passive exoskeleton, and active assistance during isolated finger flexion and object grasping tasks. The real-time performance of the deep learning-based framework will be assessed by classifying isolated finger flexion and two grasping styles, visualizing the predicted movements with a digital twin, and evaluating both classification accuracy and movement-to-visualization latency. To evaluate the framework's capacity for fine-motor control, the model will be specifically tested on its ability to differentiate between individual flexion and extension motions for all five fingers. Classification performance will be quantified by recording correct predictions, self-corrections, misclassifications and response time during the task.

We hypothesize that a EEG–EMG multimodal fusion framework will have improved motion classification accuracy when compared to a unimodal EEG or EMG framework. We further hypothesize that adding convolutional feature extraction and attention to an LSTM improves the robustness and ability to generalize compared with standalone LSTM. We hypothesize that an exoskeleton with a modular tendon-driven architecture and integrated proprioceptive sensing and tactile feedback do not significantly increase muscular effort when worn passively. Furthermore, the exoskeleton is designed to preserve natural hand dexterity, ensuring that its physical structure does not impede finger mobility.

2. Methods

2.1. Design and Fabrication of SoFiE

The exoskeleton hardware platform developed in this work, hereafter referred to as SoFiE (Soft Finger Exoskeleton for Intelligent Grasping), is a wearable assistive device designed to support grasping through flexion of the index finger and thumb. According to a review by Hill et al. a problem that exists with the development of exoskeletons today is that engineers and scientists do not have the end user in mind during development (Hill et al 2017). During the development of our proposed design, we have tried to have a potential end user in mind. The system is developed with emphasis on low profile, compliance, portability, and user comfort. Rather than using rigid revolute joints that must be precisely aligned with the biological finger joints, SoFiE relies on a compliant glove-like structure that conforms to the user's hand and follows the natural kinematics of the digits during motion. This approach reduces the need for exact anatomical alignment and allows the device to remain lightweight and unobtrusive during use.

SoFiE was first developed with only the index finger module. The index finger was chosen to prove the concept as it was simpler to actuate than the thumb, but is still an essential part in completing basic manipulation tasks. The index finger is also similar in shape and size to the three other fingers, making it possible to create those modules in the future with only slight modifications to the design of the original module. The index-only configuration will be presented first and used later for characterization of the two novel sensors developed during this work. Even though the two configurations of SoFiE share many similarities due to its modular design, the version with actively assisted index and thumb (pinch configuration) will be presented later. The pinch configuration will be used in conjunction with the deep learning-based control framework as a final configuration of SoFiE. With time, it could be expanded to assist all five digits of the hand. A view of the design featuring the index and thumb assisting version of SoFiE is shown on Figure 2, with a general overview of the entire system on Figure 2**A**.

The exoskeleton is fabricated from 3D-printed thermoplastic polyurethane (TPU). A variable-shore TPU (varioShore TPU, colorFabb) is used for the main glove structure, allowing the printed parts to combine flexibility with sufficient local stiffness for tendon guidance and load transfer. The varioShore TPU contains a foaming agent that, when activated, creates less dense parts that also have a softer texture, making it more comfortable against the skin. The foaming property is also what makes the filament able to change shore hardness during printing. It can vary from 92 Shore A to 55 Shore A from a harder TPU to a soft TPU. The foaming is controlled using the temperature at which the TPU is printed at, and for a compromise between toughness, a soft texture, and the increased flexibility that comes with the foamed filament, the varioShore TPU parts of the exoskeleton was printed at 220 °C. The glove has spring elements integrated into the wrist band to allow for further local stretching during wrist motion, letting the glove keep it's tight fit when the user moves their hand. The finger module is constructed from open ring-like segments (phalanx rings) positioned along the phalanges and connected dorsally by the stretch sensor, a compliant spring structure (see Figure 2**E**). On each side of the finger, the phalanx rings are also connected using a small bar for stabilizing the rings and keep them aligned on the finger, have the right spacing, and to reduce unwanted twisting of the module. Everything is designed as individual modules that can be fitted together by snapfit for secure mounting and easy assembly and disassembly. The glove with both thumb and index finger is shown on Figure 2**C**, with the modular architecture shown in detail on Figure 2**E**. The modular architecture allows the device to be adapted to different users and simplifies replacement or redesign of individual elements without requiring the complete glove to be re-manufactured. The compliant material also accommodates minor geometric differences between users and helps maintain comfort during repeated bending.

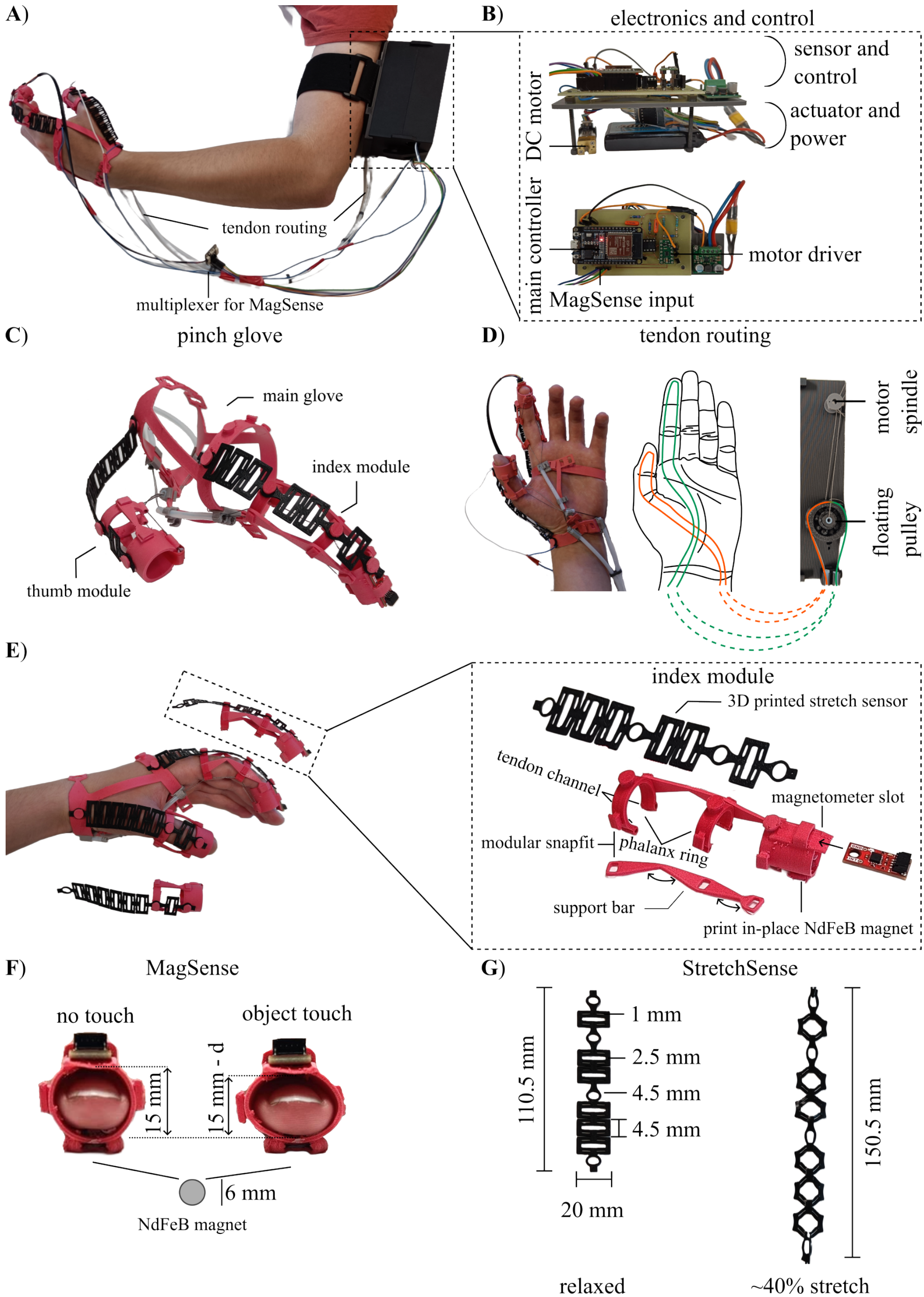


Figure 2. *Overview of SoFiE.* ***A*** *shows the exoskeleton worn by a user.* ***B*** *an interior view of the enclosure housing the electronics for controlling the exoskeleton.* ***C*** *is the main glove with the index and thumb module attached.* ***D*** *is a diagram showing how the tendons are routed for pinch motion, utilizing the floating pulley as a passive differential.* ***E*** *is closer look at how the glove is worn with a disassembled view of the index module.* ***F*** *shows the MagSense sensor doing conceptual sensing, displacing the tissue whit the distance d.* ***G*** *is the StretchSense sensor in shown in a relaxed and a fully stretched state, increasing its total length by 40%.*

To reduce the apparent weight on the hand, the actuator, battery, and control electronics are housed in a separate enclosure, which is strapped to the upper arm. Placement of the electronics can be seen on Figure 2**A** and a closer view of the electronics are found on Figure 2**B**. This arrangement moves the heaviest components closer to the body and leaves only the glove, tendons, and sensing elements on the distal limb. The resulting architecture preserves wrist mobility and reduces the amount of hardware directly mounted on the hand, which is important for both comfort and practical use. The electronics are configured in a stacked layout with two layers. On the top layer, a custom circuit-board is mounted. To this, the ESP32 30pin development board acting as the main controller is mounted together with a Pololu DRV8835 motordriver. Incoming sensorwires, and power is also distributed through this board. A step-down/step-up buck converter is also integrated into the top layer, responsible for delivering 5 V to the system. At the bottom layer is the single Pololu micro metal gear motor with 250:1 gear ratio that actuates the exoskeleton, and a 7.4 V 1000 mAh Lithium-Polymer (LiPo) battery responsible for powering the system. The 7.4 V of the battery is too high for the ESP32 and the motor, and it is therefore stepped down to 5 V using the aforementioned buck converter. The layered architecture is illustrated on Figure 2**B**.

Actuation Mechanism

Finger flexion in SoFiE is generated using a tendon-driven actuation mechanism powered by a single 250:1 micro metal gear motor located in the upper-arm enclosure. The motor rotates a small spindle that winds or unwinds a thin steel tendon. The mechanism is shown on Figure 2**D**. Mechanically, the tendon functions as a remote flexor element: when the spindle winds the tendon, the length of the tendon is reduced, thereby creating a tensile force along the routed tendon path. Because the tendon is constrained by polytetrafluoroethylene (PTFE) guide tubes along the arm and by tendon channels on the phalanx rings, this tensile force is directed toward the distal part of the digit and produces a flexion moment about the finger joints. SoFiE therefore does not actuate each joint separately using rigid links or aligned revolute joints. Instead, the tendon routing, compliant exoskeleton structure, and natural kinematic constraints of the hand together determine the resulting grasping motion. Since tendons can only transmit tensile forces, finger extension is not actively driven by the motor but is instead provided by the elastic restoring forces of the finger and the compliant structure of the glove when tendon tension is released.

In the index-only configuration, the tendon is secured to the motor spindle and routed from the upper-arm enclosure through a PTFE guide tube to the wrist. From here, it continues across the palm and along one side of the index finger through the tendon channels integrated into the phalanx rings. At the fingertip, the tendon loops around the distal fingertip structure in a dedicated channel before returning down the opposite side of the finger. It then enters a second PTFE guide tube at the wrist and returns toward the electronics enclosure, where it is fixed. The tendon routing follows the same path as shown in green on Figure 2**D**, just attached to the spindle instead of wound around the floating pulley. When the spindle winds the tendon, the loop is shortened and tension is applied along both sides of the routed path. This pulls from the distal region of the finger and draws the fingertip toward the palm, producing flexion in a way that is functionally similar to the natural flexor tendons of the hand. The PTFE guide tubes reduce friction between the remotely placed actuator and the main glove, while also constraining the tendon path so that the applied motor displacement is transferred primarily into finger flexion rather than unwanted deformation of the soft structure.

2.2. StretchSense: Integrated Compliant Conductive Sensing

StretchSense is a compliant conductive spring integrated on the dorsal side of each digit. Its dimensions and properties are shown on Figure 2**G**. Tjahyono et al. proposed a novel design for a flexible strain sensor for their hand exoskeleton. They created this sensor by coating a natural rubber (NR) substrate in polypyrrole (PPy), a conductive polymer, to create their NR/PPy structure or strain sensor, they use for position feedback (Tjahyono et al 2013).
StretchSense takes some inspiration from their work, but differentiates itself by being dual purpose as each StretchSense sensor serves two functions simultaneously. Mechanically, it provides passive extension by storing elastic energy during flexion and assisting the digit in returning toward its neutral position when tendon tension is released. Electrically, it acts as a strain-dependent resistor whose resistance changes as the spring is stretched during finger flexion. The sensor is also fabricated from conductive flexible TPU (Conductive Filaflex, Recreus) and is implemented as part of a voltage-divider circuit. By measuring the corresponding voltage change during actuation, the microcontroller can obtain a proxy measurement of digit deformation and thereby estimate the flexion state of the exoskeleton. Integrating the sensing element into a component that is already mechanically useful avoids the need for separate rigid encoders or external bend sensors on the fingers and supports the low-profile design philosophy of SoFiE. The dual-purpose function of StretchSense is particularly advantageous for a wearable hand exoskeleton. A single printed component simultaneously contributes to passive return and proprioceptive sensing, reducing the number of discrete parts required on the hand. In the initial hardware characterization (Section 4.1, StretchSense), different spring thicknesses were evaluated to determine the relationship between restoring force, deformation, and resistance change. The same sensing principle is retained in the final pinch configuration, with the additional thumb sensor enabling deformation measurements from both actuated digits.

2.3. MagSense: Integrated Tactile Perception

To provide tactile information during grasping, SoFiE incorporates a compact fingertip sensor we have chosen to call MagSense. A small neodymium magnet is embedded in the palmar fingertip structure, while a magnetometer (Qwiic Micro MMC5983MA Magnetometer, SparkFun) is positioned dorsally above the fingernail region. How it is integrated and conceptually works is shown in Figure 2**F**. The magnetometer has a measuring range of ±8 G (Gauss). When the fingertip is in contact with an object, the compliant TPU structure together with the underlying biological tissue deforms, reducing the distance between the magnet and the magnetometer with a distance d. This change in distance produces a measurable variation in magnetic flux density.
MagSense provides information that is complementary to both actuator position and StretchSense. While motor encoder data describe motion at the actuator side and StretchSense reflects digit deformation of the actuated digits, MagSense captures the local interaction between the exoskeleton and the grasped object and can in that way be used as a safety measure as well. In preliminary experiments, the sensor produced distinct responses when interacting with materials of different stiffness, demonstrating its potential for contact detection and estimation of object compliance (Section 4.1, MagSense). Within the complete SoFiE platform, MagSense therefore functions as a local tactile sensing modality that can support future adaptive grasp control and contact-aware safety mechanisms.

2.4. Two Finger Exoskeleton

As mentioned before, the final version of SoFiE utilizes a pinch-style grasp, where both the index finger and thumb are actuated. This is the version showcased in Figure 2 and it shares the same main glove structure, electronics, and the index module is also carried over. Although it shares a lot of similarities with the index finger version, it does add the thumb module and a floating pulley to the actuation mechanism which changes the tendon routing as well. The thumb module consists of two phalanx rings and a thumb specific StretchSense, but both are in a similar design to the index module.
On the present design of the index and thumb assisting SoFiE, MagSense is only implemented in the index finger module. This was chosen in order to reduce the risk of magnetic cross-talk or disturbance between adjacent sensors during grasping. Since the thumb and index finger approach one another closely during pinch motions, placing a MagSense element on both digits would cause the field of one magnet to influence the measurement of the other. By using MagSense only on the index finger, the system keeps its tactile sensing capabilities while reducing the risk of sensor interference in the two-digit architecture.

For the alteration of the actuation system, the motor still rotates the small spindle, but instead of winding the tendon acting as the primary actuation at the fingertip, it now retracts a short primary tendon that is connected to a free-floating pulley. The tendon routing and floating pulley is illustrated on Figure 2**D**. A second, continuous tendon is routed around this pulley, with one branch passing through the thumb module and the other branch passing through the index-finger module. Both branches are routed through their respective PTFE guide tubes and tendon channels before returning to the pulley, forming a closed secondary tendon loop. When the motor retracts the primary tendon, the floating pulley pulls on the secondary tendon and increases tension in both branches. This allows the single actuator to produce coordinated flexion of both the thumb and index finger. The floating pulley acts as a passive differential between the two tendon branches. If both digits move freely, the tendon displacement is shared between the thumb and index finger, producing a coordinated pinch motion. If one digit is blocked by an object or in another way have a higher resistance, further displacement is distributed to the digit with lower resistance. This allows the grasp to adapt passively to the object geometry rather than forcing identical motion in both digits. The mechanism therefore supports underactuated pinch grasping with a single motor, while keeping compliance and reducing the number of motors, drivers, and control channels needed for the wearable system. However, the differential mechanism does not allow the single motor to control each digit independently. Both tendon branches are tensioned during actuation. The floating pulley only redistributes tendon displacement between the thumb and index finger depending on the resistance encountered by each digit.

2.5. Electrophysiological Sensing Modalities

Biosignals provide a rich, continuous source of physiological sensing, reflecting both neural and muscular activity across the entire body. Nevertheless, acquiring high quality, discriminant biosignals remains a challenge due to crosstalk interference, inter-subject variability, the difficulty of selecting optimal channel locations for relevant content, and the complexities of interpreting neural perception during stimulus activation. These obstacles arise partly from the complex anatomy of the human body and the inherent limitations of current non-invasive acquisition technologies.

Electroencephalography Dynamics

Electroencephalography (EEG) is a technique for non-invasively measure the electrical dynamics of the brain. EEG signals primarily reflect the summation of postsynaptic potentials, meaning the total exchange of electrochemical signaling across the synapses, across millions of pyramidal cells. However, postsynaptic potentials do not provide explanatory power of the EEG content such as spatial, spectral and temporal features (Cohen 2017). On the contrary, specific oscillations and features have been of interest to neuroscientists for understanding perceptual, cognitive, emotional, and motor processes. Throughout this work, certain aspects have been inspected to explain the characteristics of EEG dynamics, one of which is the frequency band-limited power: referring to how the signal rhythm behaves at different frequency bands and are related to certain processes. These frequency bands are categorized as delta (0.5 – 4Hz), theta (4 – 8Hz), alpha (8 – 13Hz), beta (13 – 30Hz), and gamma (> 30 Hz). Research demonstrates that alpha and beta band suppression within the central sensorimotor region correlates with voluntary movement execution. This concerns both motor execution (ME) that refers to physical movement, and motor imagery which is the mental process of visualizing physical movement without physical execution (Ding et al 2025). Another aspect of the EEG phenomena is EEG potentials. Specific event-related potentials (ERP), are categorized by different frequency bands and cortical regions. These are distinct, time-locked reactions formed by external or internal events (Sur and Sinha 2009). For example, visual stimulation typically suppresses the occipital alpha rhythm, while performing ME reduces the mu rhythm (8 – 13 Hz), which is localized over the central sensorimotor cortex. These characteristic drops in localized frequency power are formally referred to as event-related desynchronization (ERD) phenomena (Pfurtscheller 2001). The opposite is event-related synchronization (ERS) which refers to a enhancement/rebound of the usual EEG oscillation. Such enhancements can be related to the mu rhythms during visual stimulation or a beta rebound after ME (Pfurtscheller 2001).

Electromyography Dynamics

Surface electromyography (sEMG) is a non-invasively technique in which electrodes are placed on the skin to record electrical activity provided by motor units, allowing muscle fiber contractions. The motor units are functional entities of the neuromuscular system. Each motor unit consists of a single motor neuron and all innervated muscle fibers by its axonal branches (Garcia and Vieira 2011). Motor neurons are a general term for neurons that are responsible for transmitting motor commands. The upper motor neurons are a specific type that originate in the brain and transmit signals to the lower motor neurons through the spinal cord. The lower motor neurons form the final neural pathway to the skeletal muscle fibers, where their axons branch to innervate multiple fibers (Stifani 2014). Whenever a lower motor neuron discharges, action potentials are generated at its neuromuscular junctions, that is the synapse between the lower motor neurons and muscle fibers, and the summation of these action potentials which propagate along the muscle fibers toward the tendon regions, are responsible for muscle contraction. The measured EMG activation intensity depends on the amount of active motor units and the rate at which the unit discharges, but it varies depending on the muscle group, target force and the contraction type (Garcia and Vieira 2011).

2.6. Multimodal BCI Methodology

This section presents the EEG – EMG multimodal classification pipeline used to decode hand motor intentions from synchronized neural and muscular activity. The framework combines modality-specific preprocessing with three sequential deep learning architectures of increasing complexity: a baseline long short-term memory (LSTM) network, a hierarchical convolutional extension (CNN+LSTM), and an attention-based extension (CNN+LSTM+Attention). Both unimodal classification and decision-level fusion are evaluated to examine the contribution of each modality and the potential benefit of multimodal integration. The section covers the complete pipeline shown in Figure 3, including sensor placement, preprocessing, synchronization, signal characteristics, model architectures, hyperparameter optimization, and training strategy.

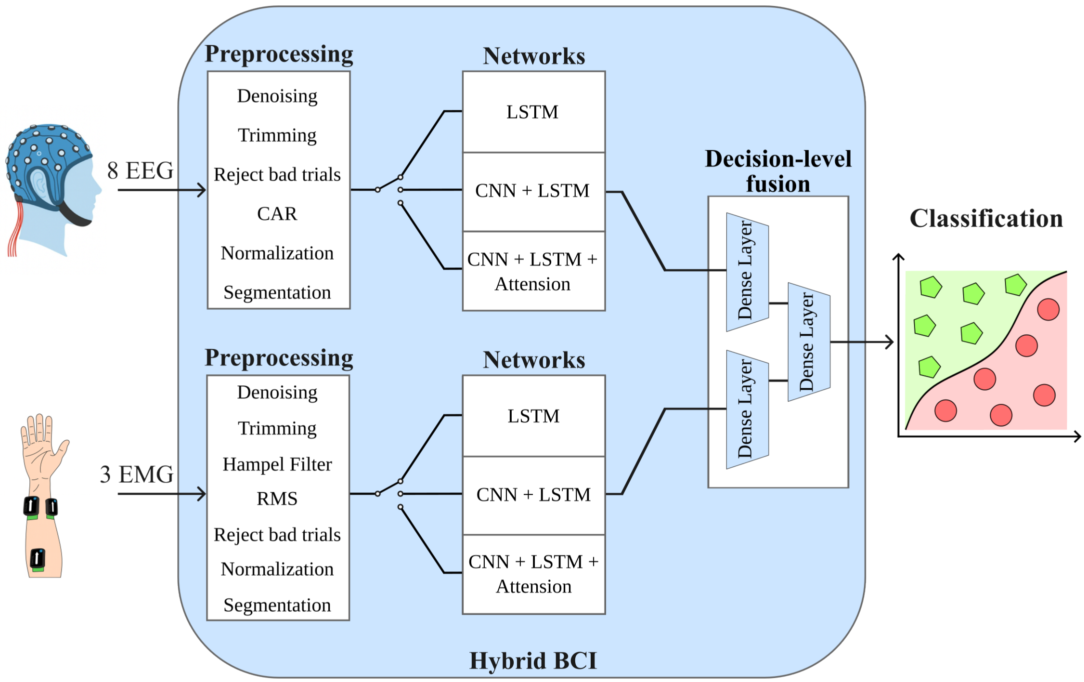


Figure 3. *The Hybrid-BCI classification pipeline, in its wholeness will present a single multimodal. Each modality are processed separately and forwarded to one of the three networks. The networks extract distinct features and forwarded to a dense layer to archive logits. These logits are provided for the decision-level fusion that evaluates the final classification.*

EEG Channel Selection and Spectral Analysis

A clear trend appears in state-of-the-art research for EEG channel selection and band frequency utilization, showcasing that the sensorimotor cortex among mu/beta bands are preferable for motor and imagery execution. The remaining section highlights which channels and frequency bands gives the strongest class separation, and for further details see the Supplementary Section S1. Five regions were defined: prefrontal (Fp1, Fp2), frontal (F7, F3, Fz, F4, F8), central (C3, Cz, C4), temporal (T3, T5, T6, T4), and partial (P3, P4), among five frequency bands delta (0.5 – 4 Hz), theta (4 – 8 Hz), alpha (8 – 13 Hz), beta (13 – 30 Hz), gamma (30 – 60 Hz). To capture the full frequency range while suppressing noise, a bandpass filter (Butterworth, 4th order, 0.5 – 60 Hz) and infinite impulse response filter (Notch, 2nd order, 50 Hz) was applied. The dataset was segmented into equal sized trials, and the resulting periods were concatenated regardless of the specific finger involved. The relative power was computed per class (distinct, predefined label that the model is trained to assign to input data) for each specific region, frequency band and participant. The results shows that relative power distribution varies across participants, on average the highest power belong to delta band at the prefrontal region, alpha band at the parietal region, and beta band at the central region. However, a repeated-measures ANOVA revealed a significant effect ($F(74, 1184) = 20.26, p \ll 0.05$), indicating differences across classes, region, and frequency band combinations. A Cohen's d analysis unfold delta, alpha, and beta in frontal and central region to be the strongest contribution of class separation (Cohen's d $\in [0.36, 0.51]$). With that in mind, those eight channels at the unfolded frequency bands provides a strong baseline for extracting meaningful, class separation content and are used in this work.

EMG Sensor Placement Strategy

In order to distinguish between all fingers during flexor motions, three EMG sensors were placed at the anterior forearm at the participant dominant hand. Figure 4**B** illustrates the EMG sensor locations on the forearm, denoted as channel 1 though 3. Channel 1 is placed at the middle of the wrist flexor group at the superficial view, primarily consists of the flexor carpi radialis, palmaris longus, and flexor carpi ulnaris, each of these muscles contribute to flexion of the wrist (Okwumabua et al 2025). Though experimental examination, this placement showed the highest activation during ring and middle finger flexor motions. Channel 2 is placed to acquire the intermediate group activation, mainly the flexior digitorum superficials. This muscle allows for flexion of the wrist, the knuckle joints, and the middle joint of our four fingers, excluding the thumb (Okwumabua et al 2025). This placement have shown superior activation during index and pinky activation. Lastly, channel 3 is placed to acquire activation from the deep view flexor pollicis longus, which highlights similar functionality as the flexior digitorum superficials, but innervate for the thumb instead (Okwumabua et al 2025).

EEG Preprocessing

The first preprocessing step is denoising the signal to suppress drift and prioritize frequency bands with discriminative information. A bandpass filter (Butterworth, 4th order, 0.5 – 30 Hz) and infinite impulse response filter (Notch, 2nd order, 50 Hz) was implemented to extract those frequencies related to ME and suppress power-line interference. The second step is trimming. The initialization of data streaming for both the Trigno and OpenBCI hardware requires a startup period for the signal to stabilize. Conversely, terminating the data stream can lead to dropped data packets. To counter these issues, a three-second baseline recording, was included at the beginning and end of the experiment. This buffer was subsequently removed from the dataset during preprocessing, preventing unstable signals, the loss of informative data packets, and edge transient distortions from filtering. After trimming the data, it is necessary to inspect if the data contain artifacts and remove them. Because normalizing immense artifact spikes, will shrink the relevant signals and lower the signal-to-noise ratio (SNR), resulting in poor data quality. The rejection routine is described later in the section (Synchronization and Rejection Techniques). The Common Average Reference (CAR) computes the average potential across all of the channels and subtracts it from each individual channel. Its a valuable technique for improving SNR and removes common-mode noise. CAR is stated to outperform other standard types of referencing, leaving more discernible signals (Ludwig et al 2009). The mathematical expression is stated in Table 1.
Finally, the EEG is normalized into a common scale using Z-score, showcasing how data samples falls relative to the mean. The normalized data is then segmented into the fixed trials, which holds the rest, onset, and offset periods.

EMG Preprocessing

In order to suppress high frequency interferences and low frequency motion artifacts. The frequency range varies across state-of-the-art studies. Based on the Trigno user guide and relative literature (Delsys, Inc. 2021; Lee et al 2022; Phinyomark et al 2010), a bandpass filter (Butterworth, 4th order, 20 – 450 Hz) and infinite impulse response filter (Notch, 2nd order, 50 Hz) was implemented for suppressing high and low frequency components and power-line interference. The filtered EMG data is trimmed exactly as EEG data when extracting the periods from the experiments. Looking back at our reasoning for the sensor placement strategy, a common trait occurs for the individual muscle groups, as they are closely interconnected. There exists a high possibility that a specific EMG channel captures neighboring muscles activity, because of the difficulty of accurately isolating the targeted muscle, which is refereed to as crosstalk (Bhowmik et al 2017). For example, a thumb contracting can be visible on channel 1, because the pollicis longus (channel 3) closely bypasses the wrist flexor group. Additionally, unintentional muscle

twitches, which are very common for people suffering from tremors, can cause immense impulsive spikes in EMG measurements. These outliers can influence the statistical parameters of EMG recordings, thus badly influence the classification model finalization.
The outliers have been analyses with an power spectral density estimate, highlighting these artifacts belong to high frequency components of approximately 180 – 220 Hz and exists within the defined EMG range of 20 – 450 Hz. To address this issue, a statistical filtering method, the Hampel filter (Bhowmik et al 2017) is implemented to repair the signal. The Hampel filter implementation is stated in Table 1. A set of local median windows are extracted from the continuous data-stream using a median filter, half the window size is defined by $k = 100$. The Median Absolute Deviation (MAD) reflects the median of absolute differences between each data sample and the corresponding median window, and is applied with a normal scaling factor (1.4826) keeping MAD to a standard deviation scale. If the absolute differences between the signal sample and corresponding typical value ($|x_i - m_i|$) exceeds the MAD criterion, that sample is replaced by its local median. The threshold factor ($\lambda = 2$) is empirically selected to suppress muscle artifacts while preserving clean signals components.
The repaired signal from Hampel filtering will be transformed from a chaotic signal into a smooth, easy-to-read waveform envelope, which quantifies the intensity and force of muscle activity. This is acquired by using Root Mean Square (RMS), the difference between the Hampel filtered signal and RMS envelope can be seen at Figure 4**C**. Achieving this goes as follows: a 1D moving average filter was adopted to extract segments of the signal. The window size hold 500 ms of the signal and increments by 50 ms (90% overlap), the RMS are computed for every window using the formula in Table 1. Lastly, the RMS envelopes are normalized using Z-score and segmented into equally sized trials, same as with the EEG preprocessing.

Synchronization and Rejection Techniques

Marker alignment is an important tasks, it ensures that each trial begins at $t = 0$ with a rest period and at exactly $t = 3$ the onset period begins, thereby preventing data drift between trials.
Data acquisition for each modality is implemented within a multi-process framework, enabling parallel execution. Upon initialization, each modality process sends an acknowledgment signal to the main process once it is fully ready. A blocking event object ensures that all modalities are synchronized and prepared before acquisition begins. During this waiting phase, the modality buffers are continuously flushed to remove unused samples and prevent stale data from affecting synchronization. Once the main process receives acknowledgments from all modality processes, the experimental protocol starts, and each process continuously acquires data without additional delay. Since trials are aligned, it is possible to interpret trials across EEG and EMG data streams as the same modality, since they explain the same event. In cases where human error or hardware malfunction occur during the experiment, they need to be addressed and removed. For example, if the participants performs an unintentional motion, like performing onset motion during the rest period, which can be observed in EMG recordings. Additionally, swallowing, jaw clenching, coughing, and head movement causes abnormal EEG signal behavior. To account for both artifacts and unintentional motions, two methods are suggested: auto-reject and manual-reject. For manual-reject scenarios where the participant performs the wrong motion, will be noted during the experiments and manually marked as a bad trial, thus removed during the artifact rejecting preprocessing step. Auto-reject requires more finesse. A channel-wise rejection threshold are computed across all trials within the same experiment. If a trial peak-to-peak value exceeds this threshold, the trial is considered bad resulting in removing this trial from all modalities. Those channels not included in the classification pipeline, does not affect the decision if the trial is bad. The mathematical expressions are stated in Table 1 (Jas et al 2017).

Table 1. *Mathematical formulation of the preprocessing pipeline, including signal processing and artifact rejection methods.*

Feature	**Formula**	**Dim.**
Z-score normalization	$z_i = \dfrac{x_i - \mu}{\sigma}$	$\mathbb{R}^{T\times Ch}$
Root Mean Square (RMS)	$RMS = \sqrt{\dfrac{1}{N}\sum_{i=1}^{N} x_i^2}$	$\mathbb{R}^{T\times Ch}$
Common Mode Average (CAR)	$CAR = x_c - \dfrac{1}{N}\sum_{i=1}^{N} x_i^2$	$\mathbb{R}^{T\times Ch}$
Hampel Filter		
Median window	$m_i = \text{median}(x_{i-k}, \ldots, x_i, \ldots, x_{i+k})$	$\mathbb{R}^{T\times Ch}$
Median Absolute Deviation (MAD)	$MAD_i = 1.4826{\cdot}\text{median}(\lvert x_j - m_i\rvert), \quad j \in [i-k, i+k]$	$\mathbb{R}^{T\times Ch}$
Outlier criterion	$x_i = \begin{cases} m_i, & \text{if } \lvert x_i - m_i\rvert > \lambda \cdot MAD_i \\ x_i, & \text{otherwise} \end{cases}$	$\mathbb{R}^{T\times Ch}$
Autoreject		
Peak-to-Peak Amplitude	$PTP = \max(X) - \min(X)$	$\mathbb{R}^{E\times Ch}$
Channel Median	$\tilde{m} = \text{median}(PTP)$	$\mathbb{R}^{Ch}$
Channel MAD	$MAD = \text{median}(\lvert PTP - \tilde{m}\rvert)$	$\mathbb{R}^{Ch}$
Rejection Threshold	$t_r = \tilde{m} + \tau \cdot MAD$	$\mathbb{R}^{Ch}$
Epoch Rejection Rule	Epoch e is bad if $\exists c \ : \ PTP > t_r$	

Notation: x_i = signal sample, x_c = signal for channel c, N = number of samples in window, k = half-window size, λ = threshold factor, τ = tolerance threshold, X = signal sequence for one trial, E = number of epochs, T = number of samples, Ch = number of channels.

Preprocessed Signal Characteristics

Figure 4 provides an overview of the biosignal responses to flexion contractions of the index finger (Figure 4**C, D, E**, left column) and the thumb (right column), averaged across 90 trials. Figure 4**B** illustrates the placement of three Delsys Trigno EMG sensors on the forearm muscles, which were empirically localized to capture individual finger activations. The rationale for this configuration is detailed in Section 2.6 (EMG Sensor Placement Strategy). Figure 4**C** displays the muscular activation at channel 2 during index flexion (left) and at channel 3 during thumb flexion (right), showing both the raw EMG sequence and its preprocessed RMS envelope for one trial. Each trial consists of three periods: resting, onset, and offset. During the resting period, the participant's hand is relaxed. The onset period denotes the initiation of voluntary movement, represented by the hand silhouette. After approximately 0.7–0.9s, the amplitude increases due to muscle contraction, peaking when the finger is fully bent.

The offset period marks the transition back to the resting state, where the envelope returns to the baseline noise floor. Figure 4**A** outlines the envelope distribution across all 90 trials for each channel and movement. Generally, the median values reside within the baseline noise floor because the trials remain in a resting state for the majority of the duration. However, the highest concentration of data (the upper quartile and whiskers) shifts to channel 2 during index finger motion and channel 3 during thumb motion. This distinct distribution indicates that these channels provide independent features for classifying or explaining the two finger movements.

Figure 4**D** visualizes the preprocessed, averaged EEG recordings during the trial periods. The signals are captured over the sensorimotor cortex within the left hemisphere (C3), as illustrated by the inset. By analyzing the time-domain signal behavior, distinct EEG phenomena, that align with supporting literature, are observable. Although, a perfect one-to-one comparison between EEG patterns remains challenging due to intra- and inter-subject variability, differences in hardware equipment, environmental influence, and variations in preprocessing techniques, all of which constrain signal quality. Among these, state-of-the-art literature indicates that approximately 15 – 30 % of users are unable to produce clear, easily detectable sensorimotor rhythms, such as mu and beta ERD. These deficiencies underscores the complexity of EEG interpretation and challenge of identifying domain-invariant EEG feature representations (DH Kim et al 2023). To overcome these signal-to-noise ratio challenges and reveal underlying cortical potentials, averaging multiple trials is a necessary approach to mitigate background noise while retaining phase- and time-locked neurological responses (Borràs et al 2022). Prior to physical movement onset (Figure 4**D**, 0 – 3 s), the cortex remains in a steady baseline oscillation state.

Following the initiation of finger flexion at 3.0s, an increase of electrical response is observed (S_1). This transient peak represents the primary motor potential (MP), which is a trigger response generated by the pyramidal neurons activations within the contralateral sensorimotor cortex (C3) during the transition to an activation state (Cui and Deecke 1999). Following this initial ME, the signal enters a transition phase (S_2) between 3.0 – 4.0 s. Here the waveform dynamics reflect a suppression of signal amplitude, driven by the occurrence of ERD in the sensorimotor mu frequency bands, due to the execution of motion. Immediately after (S_2) a signal rebound, causing an increase in waveform magnitude, can be observed in (S_3). This behavior is related to the motor event stimuli caused by reaching the full flexion position (Barry et al 2014; Pfurtscheller 2001) and settles to the baseline prior to the offset period. The offset period shows similar behavior as the onset period.

Figure 4**E** displays the topographical EEG potential mapping across the entire scalp grid. During the baseline resting state (0.0 – 3.0 s), the overall cortical topography remains neutral, with a minor positive polarization localized to the frontal and prefrontal regions for both the index finger and thumb tasks. Around the moment of ME, the negative potential emerges, localized specifically over the left sensorimotor cortex (C3), suggesting that contralateral movement activates the opposite hemisphere of the brain. (Cui and Deecke 1999). Notably, the thumb activation maps display generally lower absolute potential values compared to the index finger, suggesting variations of cortical intensity during thumb flexions. Finally, during the movement offset period, a broader spatial distribution are affected, engaging electrodes C3, C4, and Cz during the extension motion of the finger.

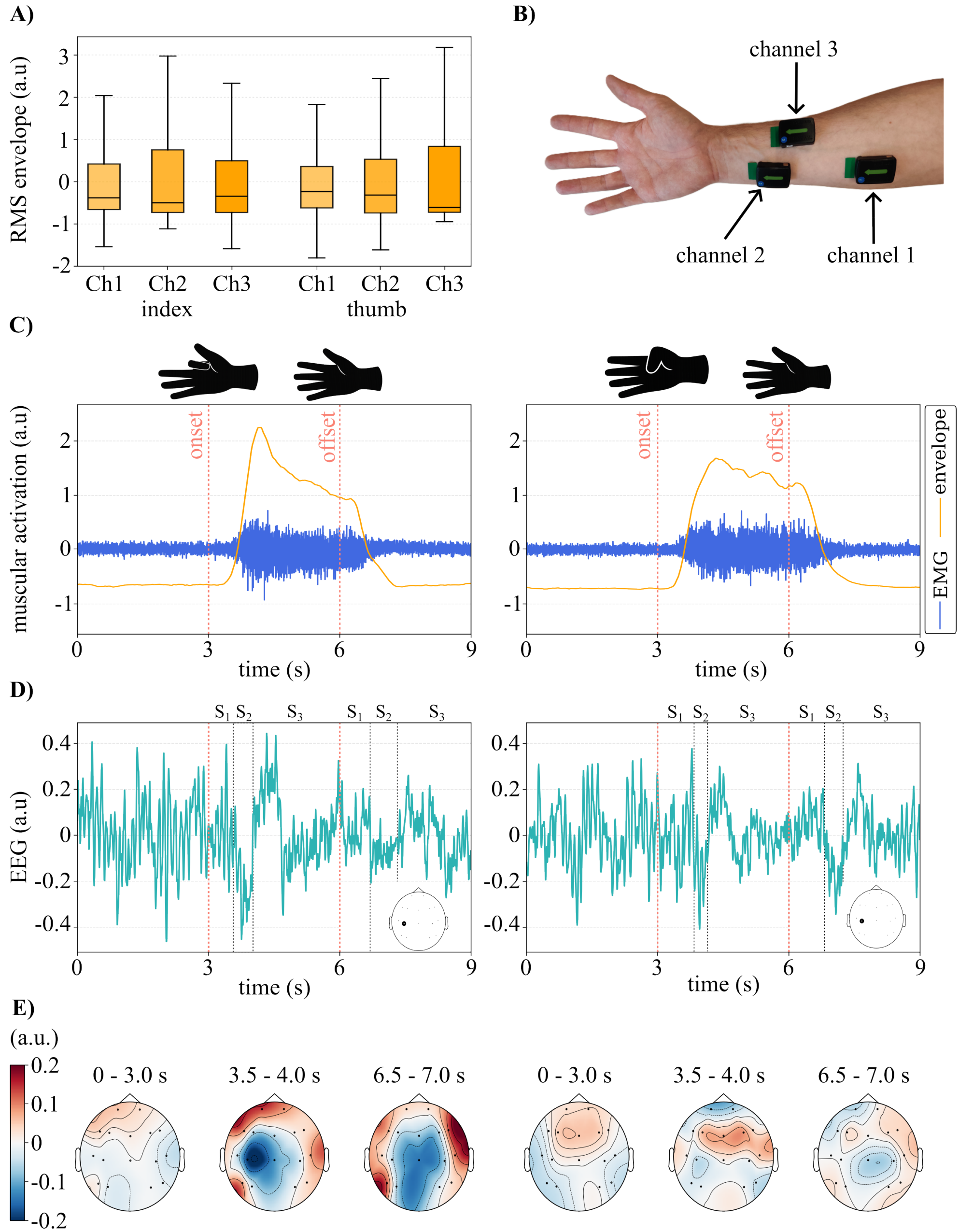


Figure 4. *A displays the envelope distribution of individual fingers at each channel.* ***B*** *shows the EMG channel locations. Channel 1 is placed on the wrist flexor group, channel 2 at the flexor digitorum superficialis, and channel 3 at the flexor pollicis longus. The remaining figures* ***C, D, E*** *displays the muscular and neural response for index finger flexion (left column) and thumb flexion (right column), illustrated by the silhouette hand.* ***C*** *shows the average raw and processed EMG (envelope).* ***D*** *is the processed EEG sequence at C3, illustrated by the inset.* ***E*** *is its corresponding topographical mapping across all channels within the time-periods.*

Classification Architecture

This section presents the classification networks used to decode ME from the EEG and EMG modalities. To investigate how the contribution of each modality can provide useful information, a set of networks are proposed to extract features from the signals. However, effectively exploiting information remains challenging particularly in subject-independent classification scenarios where inter-subject variability is high. This work proposes a set of networks with increasing complexity to exploit both temporal and spatial features among long-term dependencies. These networks are LSTM, CNN+LSTM, and CNN+LSTM with attention mechanism, throughout this work the corresponding networks will be referred to as N_1, N_2, and N_3, respectively. Each network's effectiveness will be evaluated across three modalities: EEG-only, EMG-only, and the decision-level fusion using both modalities. The documentation and approach are based on offline operation.

Long Short-Term Memory

The long short term memory network (LSTM) is a special kind of recurrent neural network capable of learning long term dependencies from sequential data. In other words, LSTM computes a feature presentations influenced by past events.

How does it work: the LSTM architecture is shown in Figure 5, if it is only LSTM-based (not influenced by other networks) the input sequences are switched to biosignals from either EEG or EMG. The input is otherwise features from a CNN. The gray enclosure box, illustrates the unrolled LSTM, which processes the first data sample x_1 to the last data sample x_T. For each data sample x_t a corresponding hidden state h_t and memory cell C_t are forwarded to the next data sequence x_{t+1}. The h_t captures short-term dependencies at current time, where C_t stores long-term dependencies and context given the conveyor belt effect of the unrolled LSTM. Both h_0 and C_0 are initialized to zero.
Additional configurations involve increasing the depth of the LSTM layers and employing a bidirectional LSTM to capture both past and future context within the input sequence. The red enclosure box, highlights one LSTM cell that incorporates a sigmoid function σ and tanh function.

The *forget gate* sum up the input sample x_t and hidden state h_{t-1} after multiplying the them with the corresponding weights and passing them through the sigmoid function. The σ decides which information to accumulate, where values of σ close to zero removes information and close to one retains it.
The *input gate* adds useful information to the memory cell. The tanh function creates a potential long-term memory, where values ranges between -1 and 1 for all possible values.
Similarly to the *forget gate* the sigmoid function regulates a percentage of the potential long-term memory for the LSTM to remember. The updated long-term memory C_{t+1} is forwarded into the next LSTM cell and passed through a tanh function to create a potential short-term memory and the percentage is retained by the sigmoid function, producing the hidden state h_t for the next LSTM cell and potentially another depth layer if enabled. The final output can either be the last hidden state h_T^1 or h_T^l , where l is the last layer depth, with the dimensionality of (B, U), where B is the batch size and U is the number of hidden units. The last hidden state serves as a condensed summary of the entire input sequence for the dense layer. The alternative is forwarding the hidden state for each input sequence $(h_0^l, h_1^l, \ldots, h_T^l)$ to the attention mechanism, with the dimensionality of (B, T, U).

Convolutional Neural Network

The one-dimensional convolutional neural network (1D CNN) is a deep learning architecture used to process one-dimensional sequence data by sliding filters across the time-domain. The 1D CNN is capable of extracting a spatial-temporal feature map from the signal.

How does it work: the CNN architecture is shown in Figure 5, where a fixed size kernel (k) slides across the time-domain until the full sequence (T) is reached, where a full sequence descries one trial period. The kernel spans across all channels and the CNN computes a weighted sum over both the time-domain and all the channels at once, resulting in spatial relationships across channels. Besides the kernel size which determines the receptive field of the spatial feature map, the CNN expand its capacity by filters (F) to identify different, unique feature in the data. An analogy could be that the kernel size (k) is the size of the looking glass, deciding how much of the input we see at once. The filters (F) are different learned lenses placed over that looking glass, each highlighting a different type of pattern. To ensure stability and balance, BatchNorm is performed to normalize across batches and forwarded into a activation function for adding non-linearity. The MaxPool applies a 1D maximum pooling operation over the sequence with a kernel size of two. This effectively downsamples the sequence dimensions by keeping the strongest activation, leading to the dimensionality of (B, T/2, F). Lastly, dropout is applied as a regularization technique to prevent overfitting, by setting elements to zero with a probability value.

Attention

There exists many versions of attention mechanisms, but in general they try to solve the long sequence problem by selectively concentrate on specific information, while ignoring irrelevant distractions. An effective and simple version was introduced in 2014, commonly known as Bahdanau attention (additive attention), utilizing encoder-decoder framework to improve sequence-to-sequence networks. Even here different variations of additive attention is introduced, but the core concept remain the same. Our work takes inspiration from a general attention pooling mechanism, which does not require a decoder (Santos et al 2016). The encoder, network flow, and components of the attention mechanism is illustrated in Figure 5.

How does it work: the input sequence is encoded by the LSTM, providing the sequence of hidden states, H. If bidirectional is enabled, twice as many hidden states are available, where forward $\overrightarrow{h}_t$ and backward $\overleftarrow{h}_t$ are concatenated for the attention block. Looking at the red enclosed box and corresponding equations in Table 2, the query (Q) represents what information we are looking for, which holds the condensed summary of the sequence. The key (K) represents the identifier of information, where value (V) contains the actual information at each sequence input. A relationship is established between Q and the information for every step K_t, which can be interpreted as: given the attention understanding from the whole sequence, which time steps are important. The alignment scores (e_t) intermediate the relationship strength, by taking the tanh function for a linear transformation of z_t, where W_a and b_a are the linear transformation weight and biases, respectively. Afterwards a dot product is computed by the learnable weight vector ($v_a^\top$), by converting it into a single score for every step. The softmax normalizes the alignment scores producing attention weights (α_t), acting like probabilities indicating importance. Finally, a context vector (c) is computed as a weighted sum between attention weights and the encoder hidden states. Allowing the network to concentrate on the most relevant temporal features within the sequence.

Table 2. Equations for the additive attention mechanism.

Step	Formula
Input sequence	$K = V = H = [h_1, h_2, \ldots, h_T]$ $K_t = V_t = h_t, \text{ for } t = 1, \ldots, T$
Query expansion	$Q = h_T$
Concatenation	$z_t = [Q; K_t]$
Alignment scores	$e_t = v_a^\top \cdot \tanh(W_a \cdot z_t + b_a)$
Attention weights	$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)}$
Context vector (weighted sum)	$c_t = \sum_{t=1}^{T} \alpha_t V_t$

Notation: h_t is the hidden state at time step t, T is the sequence length, v_a, W_a, and b_a are learnable attention parameters, α_t is the normalized attention weight, and c is the context vector.

The dense layer advance from features to classifications. See the last block element in Figure 5. The input features are either the last hidden state from LSTM or the context from attention. Each dense layers consists of two linear layers, an activation function, and a dropout layer. The primary objective of this configuration is to reduce feature dimensionality prior to classification. The output z consists of logits, which are unnormalized and unbounded scores. The class with the highest logit value represents the model's most confident classification for the input biosignal. When using both modalities, the original architecture generates z_i and a mirrored version generates z_j where they are merged afterwards into the final layer. This convention of consolidating multi-modal logits and mapping them into a new dimensional space, is called decision-level fusion. It allows individual modalities to interpret the biosignal and predict the corresponding task, whereas the decision-level fusion concludes the final outcome based on the classification of both modalities (Leeb et al 2011). For example, the network decodes with high confidence that the user executed a prehensile motion given the EEG response, however the mirrored network of EMG decodes correctly which muscle activations are involved although it is confused whether the motion is grasping around an object or releasing the object.

The final classification outcome looks at the full picture, weighing the confidences and conclude the task and muscle group involved. Another example, involves decoding of muscle artifacts, like muscle flicks, which might bypass the preprocessing routine causing the EMG model to have a low confidence classification of voluntary contraction, however if the EEG model counteract it by saying that it is in idle state with a high confidence, the decision-level fusion leans toward ignoring the activity from the muscle flick. Note, that the concatenation of z_i and z_j are normalized using z-score to ensure the logits can be combined fairly, since individually, the confidence scale and magnitudes can appear differently. Otherwise, one model with large-scale logits can dominate a more accurate model, which produces smaller logit values.

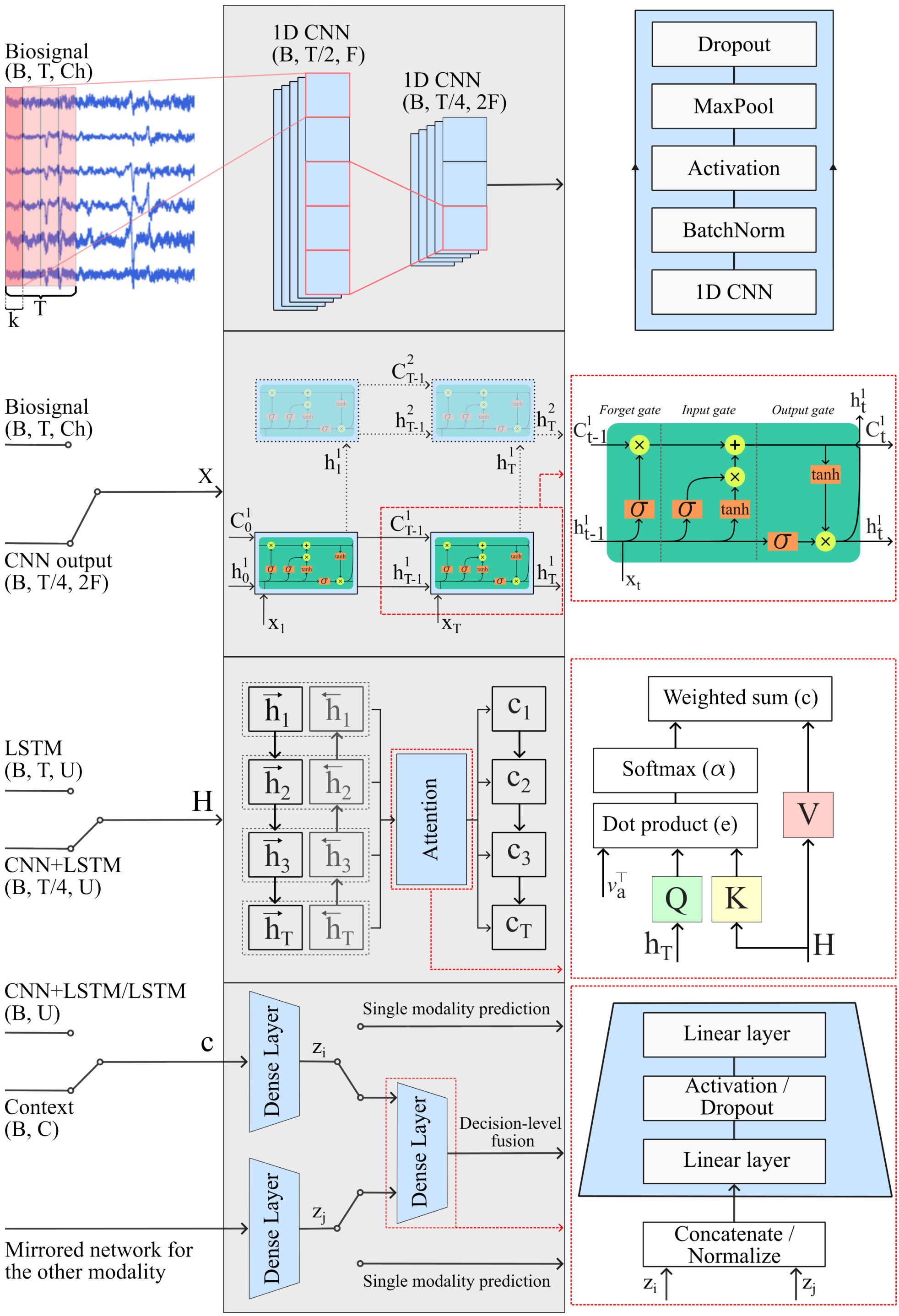


Figure 5. *Left window shows the input sequence along dimensionality, where: B: batch size, T: sequential data samples, Ch: sensor channels, F: CNN filters, D: bidirectional scalar, and U: hidden units. The middle grey boxes are the network flow from input to output. The right window highlight components within the network.*

Hyperparameter Optimization and Training Strategy

Arbitrarily choosing the best model configuration settings is nearly an impossible task and leads to suboptimal results, because the configuration varies depending on the dataset and networks. Instead, Bayesian search is recommended for hyperparameter optimization Lashgari et al 2021. It explores the hyperparameter search space, learning a distribution of hyperparameters, which provides the lowest validation loss and exploits those ranges to optimize the model configuration. Table 3 shows the search space of interest. The LSTM utilizes the hidden units, bidirectional, and layers. The CNN includes kernel size and filters. Additional hyperparameter are dedicated to the optimizer (AdamW) and dense layers.

Table 3. *Hyperparameter search space.*

Hyperparameter	**Range**	**Type**
Learning rate	$(10^{-5}, 10^{-3})$	Continuous
Weight decay	$(10^{-5}, 10^{-3})$	Continuous
Dropout	(0.1, 0.4)	Continuous
Batch size	(16, 32, 64, 128)	Ordinal
Dense ratio	(0.25, 0.5, 0.75, 1.0)	Ordinal
Hidden units	(32, 64, 128, 256)	Ordinal
Filters	(16, 32, 64)	Ordinal
Kernel size	(3, 5, 7, 9, 11)	Ordinal
Activation	(ReLU, ELU)	Choice
Bidirectional	(False, True)	Choice
Layers	(1, 2, 3)	Choice

For each model architecture, the search algorithm ran for 100 different hyperparameter settings. Each configuration were trained for 250 epochs using an early stopping condition, where the criteria for early stopping was decided to be when 25 continuous epochs did not improve validation loss.

All models were trained using a version of the train-validate-test split strategy for both subject-dependent and subject-independent classification. Subject-dependent classification can be seen as a tailor-made network to a single user, highlighting the models ability to learn individual patterns, and excels when inter-subject variability is an issue. The partitioning split ratio was set to 70/15/15% for the whole dataset.
In contrast, subject-independent classification focuses on general patterns from multiple subjects, highlighting the models robustness and ability to generalize across unseen subjects. Given the 17 participants, the partitioning split ratio was set to 14/2/1 subjects. This strategy is also called leave-one-subject-out.

Regardless of the train-validate-test split version, the biosignal dataset structure had the dimensionality of (B, T, Ch). Although the sequential data samples (T) varies depending on the modality, a prefixed window length of 3 s, corresponding to a full trial period, is chosen for the offline inference model to allow for richer context. This serves to set a baseline for the ideal condition, allowing the model to capture the transition patterns and channel relationships before evaluating the system under the stricter operational constraints required for real-time deployment.

To validate performance across all subjects, the split strategy incorporates a K-fold cross validation, where $K = 8$. For every set of hyperparameter settings, eight folds of training of completely new models are executed. For each fold, two random participants are allocated to the validation set. Once all folds are complete, a final model is trained on all participants except for one held-out individual, who serves as the independent test set.

During the training procedure, precautions against data leakage must be addressed. Data leakage occurs when information from outside the training dataset is accidentally used to create the model, or when features of the validation and test splits leak into the training pipeline. The effect is silent and causes models to present accurate, but generalize worse. Two data leakage trends was discovered during the development. For subject-dependent, the three periods within one trial needs to be grouped together, ensuring no leakage of trial-level information appear in one of the other splits (Lemoine et al 2023). The other trend is subject-level leakage and was avoided by dividing subjects to a specific split (Brookshire et al 2024).

Training strategy: For the first 75 iterations, hyperparameters are randomly sampled before optimization begins. Each configuration is trained for up to 250 epochs or until early stopping is triggered, with validation performed after every epoch. During training, a forward pass is executed each epoch, passing the biosignals through the network and dense layers to produce logits.

Afterwards, a loss function (cross-entropy loss) is executed to measure the error across predicted probabilities and true labels. The loss assists the model to become more confident by rewarding correct answers and penalizing wrong answers. A lower loss value suggests that the model assigns a high probability to the correct class and low probabilities to the incorrect classes. The loss is used to back propagate though the network leading the a gradient, which determine how weights and biases in the network contribute to the error, and which adjustments are needed to reduce it. An analogy, is that the loss function presents a curvature/landscape, where we want to descend to lower levels of the landscape to acquire a lower loss, and the gradient is our map showing which direction to travel.

Updating the weights and biases are done by an optimizer, which interpret the gradient attributes and adjusts the weights and biases accordingly. The optimizer used is the adaptive moment estimation with weight decay (AdamW), due to its properties of momentum and weight decay, it have shown promising results compared to other methods, like stochastic gradient descent with momentum and Adam (Loshchilov and Hutter 2017). Momentum ensures accelerating and maintenance of the gradient descent process resulting in faster convergence and exits from saddle points, local minima and local maxima in the curvature. The weight decay property ensure better generalization, by decouple weights from the gradient update step, it penalizes large weights to also relay on smaller weights to capture underlying patterns instead of focusing on specific dominant patterns, which can cause overfit.

Transition to Real-Time Model Deployment

Once the ideal offline inference conditions, network architectures, and modality performance are established in Section 4.3 (Subject-Independent Classification), the top-performing configuration is selected for further evaluation. This architecture and modality combination is then integrated into an online inference framework and the performance will be detailed in Section 4.4, which showcase the real-time model deployment and scalability, and Section 4.6 ultimately demonstrates the model's generalization and capability for exoskeleton control. The remainder of this section details the key component modifications required to transition the system from offline to online inference. Firstly, the training strategy must align with the operational properties of real-time inference. In other words, the sequence length used during model training must match the live data-stream inputs. This design choice represents a trade-off between temporal context and system latency. Through empirical and rational reasoning, a historical buffer containing 500 ms past information, was selected as a feasible balance for accurate classification. Coupled with an inference interval of 100 ms, new data samples are gathered every 100 ms to trigger a new classification. This is achieved using a sliding window that increments every 100 ms, creating an overlap between past and present data. Therefore, during dataset preparation for model training, the data is segmented using this sliding window approach and processed via the pipeline detailed in Section 2.6, bypassing the standard 3 s trial-period segmentation.

Segmenting the original 3 s trial period using this sliding window yields a total of N_w windows, calculated according to Equation 2.1:

$$N_w = \frac{N - W}{S} + 1 \tag{2.1}$$

where N is the total number of samples within the 3 s trial, W is the window size (corresponding to 500 ms of samples), and S is the step size (corresponding to 100 ms of samples).
Each generated window must be assigned a ground-truth class label. While windows entirely within the rest period were uniformly labeled as 'rest', windows spanning the onset and offset boundaries presented a risk of label noise. Due to participant reaction time, the early portion of a movement trial physically resembles a resting state, and a similar transition occurs during the offset period (see Figure 4**C**). To mitigate mislabeling, an automated labeling threshold was implemented. Windows within the onset period were only labeled as 'onset' once the EMG signal's window mean positive slope value exceeded the baseline noise floor calculated from the rest period. Conversely, during the offset period, windows were labeled as 'rest' once the negative slope returned to the baseline mean value.

The training strategy remains unchanged and follows the subject-dependent partitioning split ratio of 70/15/15 %.
Figure 6 displays the model inference pipeline. Data is acquired every 100 ms and transferred into a circular buffer, displacing the oldest samples. Once the buffer is fully initialized with 500 ms of historical data, the data vector is extracted and preprocessed. The deployed model processes this input to perform classification, forwarding the class with the highest logit value to majority voting.

During real-time system evaluation, the raw model classifications tended to oscillate between 'onset' and 'offset' states every 100 – 300ms once a full flexion posture was reached. This oscillatory behavior (visible in Figure 4**C**, left, between 3 – 6s) was likely caused by variations in sensor placement and the physical force applied during sustained flexion, causing the model to misinterpret descending signal slopes as intentional extensions. Without further logic, the exoskeleton would continuously contract and release the mechanical fingers in response to this instability. To mitigate this, a post-processing stage incorporating majority voting, priority scaling, and confidence rejection was implemented between the model output and the exoskeleton controller. This logic collect the five most recent model classifications into a secondary circular buffer. For each new inference cycle, the majority class is determined. If the average model confidence for this majority class exceeds a 90% threshold, the decision is forwarded to the exoskeleton, otherwise the classification is rejected. Furthermore, to handle directional transitions smoothly, whenever a flexion class ('onset') is active, its corresponding extension class ('offset') is scaled in selection priority by a factor of two, ensuring rapid, stable transitions when the user intends to extend that specific finger.

2.7. Control Architecture

SoFiE and the model inference pipeline are implemented as two separate modules that can communicate and enables the model to control the exoskeleton, as illustrated in Figure 6. The complete system is arranged using a TCP client/server architecture, where the model inference pipeline acts as the client and the exoskeleton acts as the server. On the client side, biosignals are acquired, stored in a circular buffer, preprocessed, and classified by the model. A majority-voting step is then used to stabilize the classified motion before an actuation command is sent to the exoskeleton. During operation, the client also receives the current system state and sensor feedback from SoFiE, allowing the inference pipeline to be aware of the physical state of the device, keeping it from overwriting an ongoing motion. On the exoskeleton side, the ESP32 microcontroller handles communication and interfaces with the motor, sensors, and supporting electronics that make up the SoFiE control unit. When the microcontroller receives an actuation command, it checks the internal system state and, if ready, applies the desired actuation to the micro DC motor, which retracts or releases the primary tendon through the spindle. To ensure that the motor reaches the desired position, the integrated motor encoder is used to track the current motor position. This also enables a software limit to keep the motor within a predefined range and prevent over-tightening of the tendons, which could otherwise cause discomfort for the user. MagSense is continuously sampled during operation, providing tactile feedback from the index fingertip to detect contact and infer interaction with external objects. A threshold is implemented such that actuation is stopped when the MagSense signal exceeds a predefined value, indicating contact with a solid object or a sufficiently stable grasp. StretchSense is also sampled during actuation; however, its signal is not used directly for real-time control in the present implementation, but is instead used after operation to evaluate the deformation of the actuated digit. During runtime, SoFiE sends the captured sensor data and system state back to the client, allowing the model inference pipeline to monitor the behavior of the exoskeleton. The exoskeleton currently uses a variation of a bang-bang controller, where the input is the actuation command from the model inference pipeline and the control target is either the motor position or the MagSense threshold. This controller provides a simple and robust first step for proof-of-concept validation, while alternative control strategies will be discussed in Section 5 (Latency and Controller Logic). Keeping the model inference pipeline and the exoskeleton as independent modular units allows each system to be developed, tested, and validated separately before being integrated into the complete control architecture.

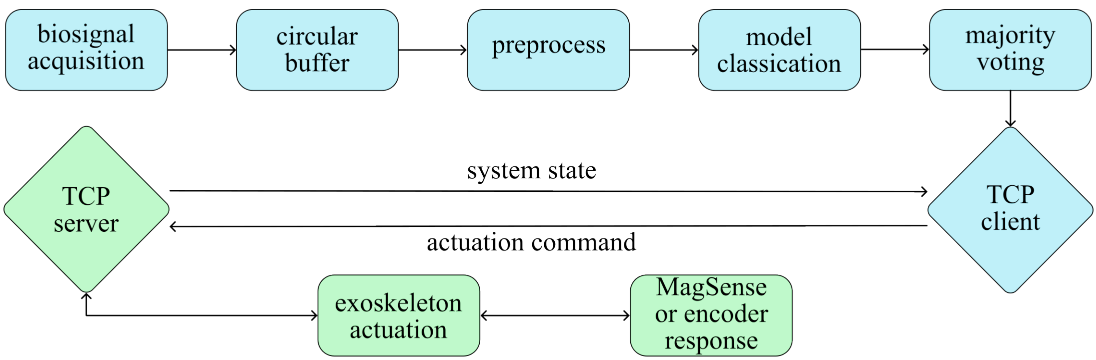


Figure 6. *System diagram of the TCP-based control architecture. The client acquires and preprocesses biosignals, performs model inference, stabilizes the output using majority voting, and sends actuation commands to the server (SoFiE). The server receives the commands, actuates the exoskeleton, monitors MagSense and encoder feedback, and returns the current sensor data and system state to the client.*

3. Experimental Methodology

This section describes the experimental procedures used to evaluate the proposed system. The experiments were designed to characterize the integrated exoskeleton sensors, assess the grasping capability of the mechanical platform, evaluate the biosignal classification pipeline, and demonstrate real-time control of the exoskeleton. First, StretchSense and MagSense were tested independently to examine their mechanical and sensing behavior. The index-only exoskeleton configuration was then used to evaluate sensor responses during grasping of everyday objects. Finally, EEG and EMG data were collected to train and evaluate the classification models, followed by real-time demonstrations of hand-motion classification, muscular-effort reduction, and model-based control of the index and thumb assisting exoskeleton.

3.1. Characterization of Integrated Exoskeleton Sensors

For the characterization of the integrated sensors (StretchSense and MagSense) the version of SoFiE with only the index finger module was used.

StretchSense

To characterize the StretchSense sensor and find the optimal thickness, five different thicknesses of the same design was printed and tested. The thickness started at 0.5 mm and incremented in steps of 0.5 mm up to 2.5 mm. Each spring was printed on a Prusa MK3s with a layer-height of 0.1 mm for better height accuracy and layer-adhesion. The filament handling and printing parameters were otherwise kept to what was suggested by Recreus, the maker of the conductive filament used. The springs were tested using a universal testing machine (EZ Test, Shimadzu). The universal testing machine, is able to either press or pull a specimen with high displacement accuracy. For the tests of StretchSense, the maximum change in displacement was measured on the subject's hand on which the sensor was meant to be used. Measuring the distance from the metacarpophalangeal (MCP) joint to the tip of the distal phalanx (the fingertip), when the index finger was fully stretched and when fully contracted, yielded a displacement on the dorsal side of 40 mm. This was therefore chosen as the maximum distance that the spring was stretched during testing.

On the universal testing machine, a 100 N load cell was rigidly mounted on the top arm, for high precision force measurements, later used to produce a stretch-strain curve and to see the relationship between force, stroke and resistance of the sensor, which will be shown in Section 4.1 (StretchSense). To measure how the electrical resistance of the spring changed during stretching and relaxation, an LCR meter (LCR-6100) from RS PRO was interfaced with the ends of the spring via a dedicated test fixture. In each end of the spring a wire was inserted during printing to securely fasten it to the spring. The test fixture then with alligator clips, attached to these wires to measure the resistance change.
An illustration of the protocol can be seen on Figure 7**A**. StretchSense was mounted in a 3D-printed fixture designed for the universal testing machine and then pre-tensioned to 0.15 N to remove any slack in the beginning, the position and force were then zeroed at this point. The universal testing machine was then set to do 31 cycles where each cycle consisted of going from 0 mm to 40 mm, then back to 0 mm. A 5 s hold was set at the zero position at the end of each cycle to allow for the material to contract. The springs were cycled at a constant rate of $1\ \mathrm{mm} \cdot \mathrm{s}^{-1}$.

Data acquisition from the universal testing machine was done through a National Instruments Data Acquisition unit (NI DAQ), which captured the analog force and displacement signals and converted them to digital values. These data streams, along with the digital resistance reading from the LCR meter, were processed and synchronized in real-time using a custom MATLAB script. For post

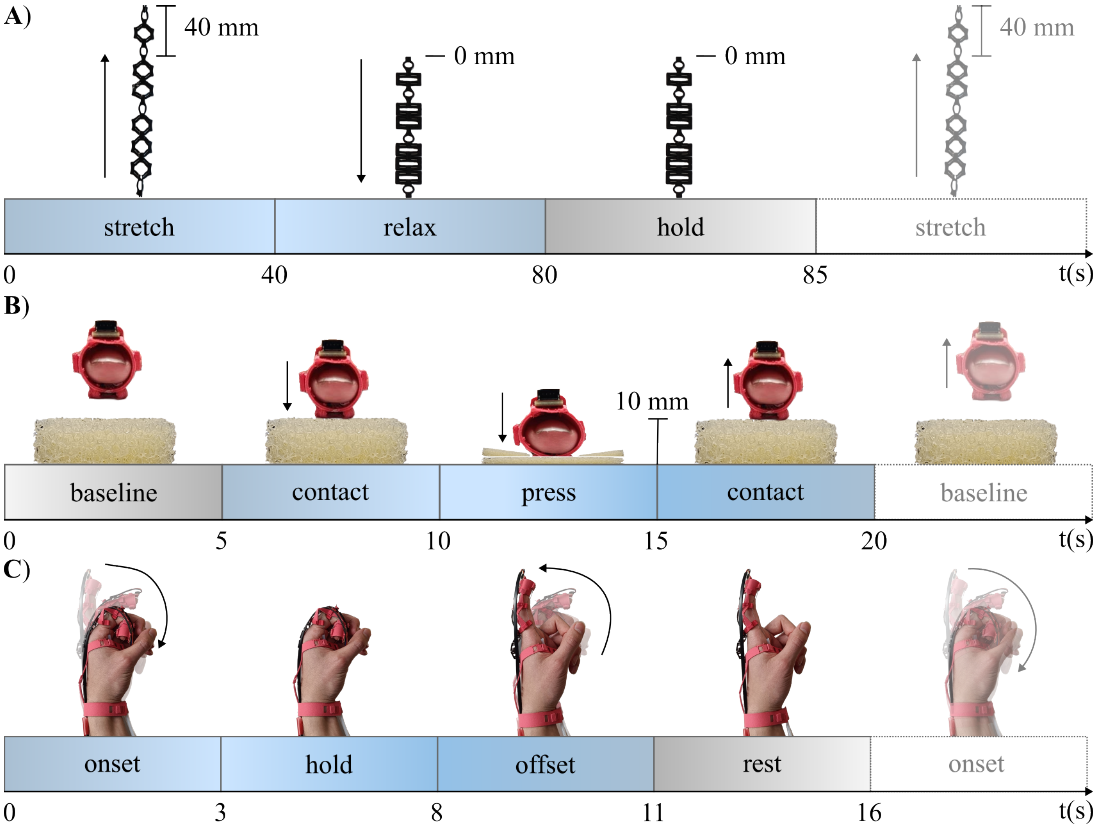


Figure 7. *A illustrates the experimental protocol for a single cycle of StretchSense on the universal testing machine with timestamps and end position.* ***B*** *illustrates the protocol for a single trial of the MagSense characterization. MagSense is held away from the foam, then placed to get the contact data, then pressed* 10 mm *into the foam and held there, then relaxed back to contact, and lastly returned to baseline.* ***C*** *illustrates how the relationship between StretchSense and the motor position was tested using the active exoskeleton, with onset, being contraction of the index finger, and offset being relaxing and returning to rest position.*

processing, the first cycle of each test was discarded as a "burn-in" outlier to account for initial material settling. The remaining 30 cycles were averaged to determine the mean and standard deviation of the force and resistance as a function of displacement.
The spring chosen for final integration was subjected to an extended test, to asses its long-term performance and transient behavior. This test followed the same protocol as before, but was extended to 501 cycles instead of 31. The experimental setup can be seen in Supplementary Information Figure S2.

Actuator Response with StretchSense

Figure 7**C** illustrates a trial for testing the motor position and StretchSense relationship. The motor was set to actuate to the exoskeletons maximum contraction of 3 rotations or ~19 rad. The encoder position and the voltage shift caused by the resistance change of StretchSense was recorded during 10 trials of 16 s each. Each trial consisted of an onset period, where the exoskeleton actuated the index finger to its maximum contraction, then the exoskeleton held its position for 5 s, allowing the measurements to stabilize. Then a offset period where the exoskeleton released the tension of the tendons, allowing the index finger to return to its rest position, which was then followed by 5 s of rest to again, let the signals stabilize. The encoder position was converted to a rotation in radians using Equation 3.1:

$$P_{\text{rad}} = C_{encoder} \cdot \frac{2 \cdot \pi}{C_{PR} \cdot GearRatio}, \tag{3.1}$$

where the position in radians P_{rad} is the current counted ticks of the encoder $C_{encoder}$ multiplied with a fraction, that converts from encoder ticks to a rotation in radians. Since the encoder measures motor rotations rather than output-shaft rotations, the gear ratio is used to scale the encoder output. C_{PR} is the amount of counts or per rotation of the motor also known as the the encoder resolution. For the Pololu motor used in SoFiE, C_{PR} is 12, as there is 6 rising and falling edges in the signal from both encoder pin A and pin B per rotation of the motor. The voltage from StretchSense is converted to a resistance using Equation 3.2:

$$R_{StretchSense} = \frac{R_1 \cdot (V_s - V_{measured})}{V_{measured}} \tag{3.2}$$

where the source voltage (V_s) is 3.3 V supplied by the ESP32. The known resistor R_1 is 1 kΩ. The voltage is measured using an ADC pin on the ESP32 with 12 bits of resolution. The voltage is filtered using a 4th order butterworth low-pass filter with a cutoff frequency of 1 Hz to isolate the low frequency changes caused by moving the finger, instead of high frequency electrical noise.

MagSense

Characterization of MagSense was done by testing against three Polyuethane (PU) foams of different densities or pores per inch (PPI). The three foams were cut into 30 mm by 30 mm squares with a thickness of 20 mm and placed in a 3D printed test fixture that both held the foams and acted as a mechanical stop for the test. The test was done by following a five step protocol (which is also illustrated on Figure 7**B**) with a total time of 20 s with each step being 5 s. First, a baseline of 5 s where the user is just hovering over the foam, then 5 s of contact with the foam where the MagSense is placed on the foam. The next 5 s, is where MagSense is pressed 10 mm into the foam and held there using the test fixture as a mechanical stop for the finger to rest against to ensure equal compression for all the tests. Then the compression is released and the sensor is brought back to contact with the foam and held here for 5 s. At the end, the sensor is lifted back to baseline. The test were repeated 10 times for each PPI. Afterwards, the measurements were filtered using a 4th order butterworth filter, removing the high frequency noise, after which they were split into trials and baseline corrected.

3.2. Object Grasping using Index Module

To demonstrate how the StretchSense and MagSense sensors respond during assisted and passive grasping, five demonstrations were performed with five different everyday objects. The five objects were a 500 mL water bottle, pencil, sponge, screwdriver and a pouch filled with pencils. These objects represent a range of physical properties. Some are rigid and thin while others are bulky or soft, demanding different degrees of index finger flexion to achieve a stable grasp. For the demonstrations, the objects were placed on a flat and stationary surface and the grasp performed was a pinch between the thumb and the index finger. The motor was set to run until it either hit its rotation limit of about 19 rad, or MagSense reports a higher value than 5 G. Meanwhile, StretchSense is read and converted to relative change to see if there is a difference in travel of the index finger between the different objects. This demonstration follows the same protocol as the actuator response test protocol mentioned earlier.

3.3. Index and Thumb Flexion Classification

A total of 17 voluntary participants agreed to assists for experiments. 16 being male and one being female in an age range between 23 – 43 (average 28 ± 5.5) years old and all right handed. Non of the participants suffered from neurological, musculoskeletal disorders nor severe hand injuries affecting their activities of daily living (ADL). However, three participants introduced a specific behavior during finger contractions, where flexing the thumb would lead to a flex of the index finger. All participants signed a consent declaration agreeing to the GDPR Article 6(1)(a) and Article 9(2)(a), given explicit consent to the processing of personal data and their personal data is anonymous and shareable for third parties for research purposes. The experimental setup, outlining hardware components is accessible in Supplementary Figure S3, and the corresponding experimental protocol can be watched in Supplementary Video S3.

The hardware for collecting EEG is the OpenBCI Cyton-Daisy board. The Cyton board communicates via a USB dongle using RFDuino radio module and has a ADS1299 analog-to-digial converter with a built-in programmable 24x gain amplifier. The wireless communication allows a sampling rate of 125 Hz. While integrating the Daisy module expands the amount of channels from eight to 16. OpenBCI is open-source and provide a graphical user interface for real-time monitoring of EEG signals, impedance control and many other widgets. The EEG cap is the Gel Electrode Cap, allowing up to 19 channels with patented, sintered Ag/AgCl electrodes. The electrodes placement follows the 10-20 system, which refers to the distances between adjacent electrodes, which ensures a standardized international method of electrode placement.
The hardware for EMG data acquisition is the Trigno wireless biofeedback base station which provides a high speed USB communication and holds up to 16 Trigno Avanti^{TM} EMG sensors. Each Trigno sensor provides a dual-mode BLE-base communication, 20 m transmission range, data synchronization between sensors, built-in nine axis inertial measurement unit, and a sampling rate of 2000 Hz.

Prior to the experiments, the voluntary participant was seated in a comfortable chair and had the EEG cap placed on their scalp. Each active electrodes were applied with conductive electro cap gel (Electrode Cap Gel, OpenBCI) using a syringe. A swirl motion was performed with the syringe to move hair aside and small amounts of gel was injected till the gel was visually connected between the scalp and electrode. Approximately, half an hour was allocated to setup before analyzing the impedance levels using the OpenBCI GUI. The waiting time was found to be important for the gel to increase in temperature, making it less viscous and able to spread properly. After the waiting time, each electrode impedance level was inspected, aiming for a value below 10 kΩ to acquire less noisy measurements. If an electrode did not reach the desired target, the electrode was further adjusted by holding it and making slight rotational motions or injecting more gel in a swirl motion. This was mostly achievable when the headset was tightly fit to the scalp, but participants with smaller head sizes, experienced less contact between the scalp and the electrodes, resulting in higher impedance levels of ~30 kΩ.

In the meantime after applying the electrode gel, 70% ethanol wipes (Abena Surface Disinfection Wipes, ABENA) was used to clean the forearm and Trigno Avanti^{TM} EMG sensors from natural oil. Double sided stickers was placed on the EMG sensors and placed on the participant forearm. The muscle locations were empirically located by advising the participant to perform contractions of individual fingers. Firstly, the participant bend sequentially between middle and ring finger, while the researcher tried to locate (feel of touch) the activated muscle at the wrist flexor group.

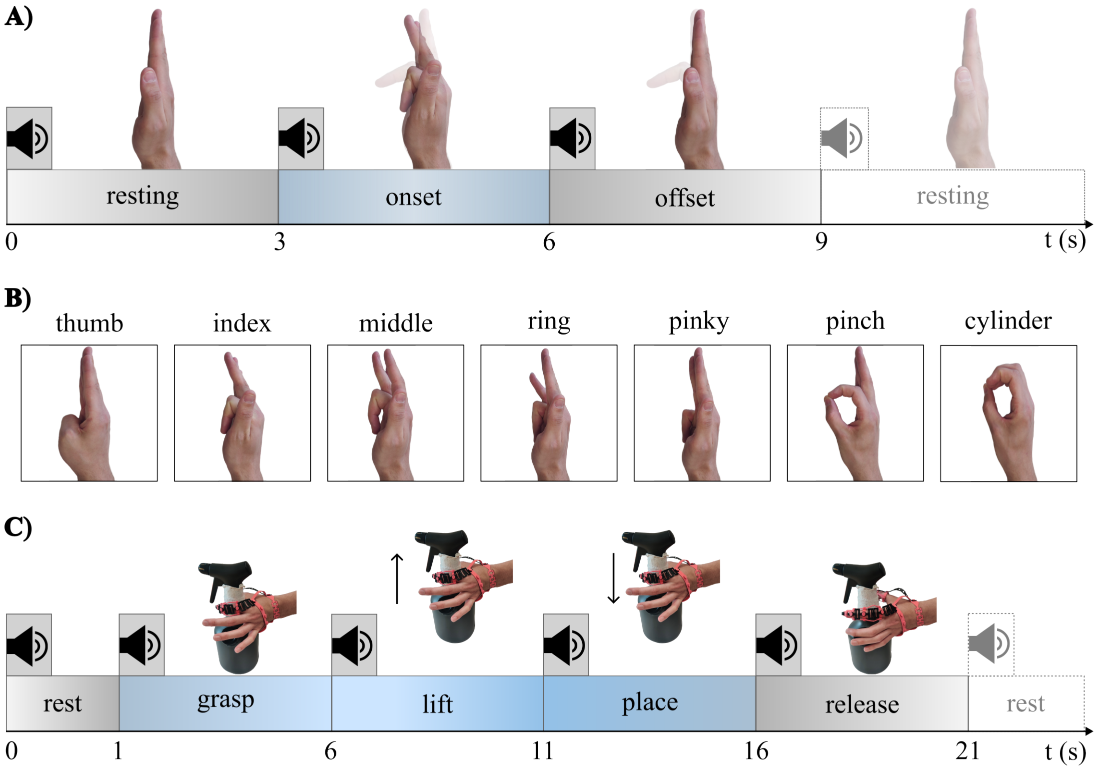


Figure 8. ***A*** *shows the single-trial experimental protocol with the corresponding task periods and motion diagrams.* ***B*** *shows the flexion motions used for the real-time hand motion classification experiment, following the same protocol as in* ***A****.* ***C*** *shows the object-grasping protocol used for muscular effort evaluation, including the five task periods, time intervals, and executed motion.*

Same procedure is carried out for the channel 2 at the flexior digitorum superficials muscle by bending the index finger and pinky, and channel 3 at the flexor pollicis longus by during thumb contractions. Thereafter, each channel are monitored in real-time while the participant performs individual finger contractions. The channel location was adjusted if the baseline peak-to-peak amplitude exceeded 20 mV. Potential causes in decreased SNR is observed to be body hair, natural body oils and stickers not fully in contact with the skin.

After all sensors were installed and verified, the participant was introduced to the experimental protocol and how to execute the desired motion. The participant places their wrist and elbow on two 3D printed varioShore TPU pillows, with the thumb pointing upwards. The participant practiced the motion to get familiar with the protocol and ensure consistent motion. The participant was instructed to close their eyes during the experiments, remain motionless, avoid swallowing, and concentrate on the execution of the motion to the best of their ability. This was to create the best possible test scenario, and avoid external artifacts during data acquisition.

Each participant went though three sessions of experiments per finger motion. One session contain 30 trials lasting nine second each. One trial is equally divided into three periods, which are rest, onset, and offset periods, see Figure 8**A**. This Figure also displays the index finger motion pattern. The beginning of each period, a voice recording was played giving a command of what the participant should do. For example, at $t = 0$ the voice said 'rest', at $t = 3$ 'contract', and $t = 6$ 'release', which repeated during the whole session. In the instance where the finger was fully bend, a gentle press was carried out on the palm to capture a higher muscular activation. The participant initialize with index finger motions for the first three sessions and afterwards perform the same paradigm with the thumb, with a break between sessions. This adds op to 90 trials per finger.

3.4. Demonstration of Real-Time Hand Motion Classification

This experiment further investigates the best-performing system (the EMG CNN+LSTM architecture detailed in Section 4.3) to assess its ability to scale to a higher number of distinct classes. Additionally, a real-time demonstration showcases the model's performance. In contrast to the previous index and thumb flexion classification, this experiment session included all five consecutive finger along pinch and cylinder grasps (see Figure 8**B**). This experiment follows the exact protocol, trials count, and motion patterns described in Section 3.3. However, the key difference are that the model was trained on a single participant and EEG acquisition was excluded. The demonstration is assessable in Supplementary Video S4, demonstrating the model's ability to recognize both motion type and active finger. A total of ten ME trials were performed per motion type, and the corresponding correct and incorrect classifications were recorded. The model's inference outcomes are forwarded to a MuJoCo simulation acting as a virtual twin. The MuJoCo environment is designed as a finite-state machine to switch between states depending on the model's real-time output.

3.5. EMG-based Evaluation of Hand Exoskeleton Assistance and Muscle Effort

This section investigates muscle effort during hand movements performed both with and without the exoskeleton. The experiments aim to characterize how the exoskeleton influences muscular demand during flexion and grasping tasks. To achieve this, an EMG-derived effort metric is compared across three operating conditions: without the exoskeleton (n_e), with passive exoskeleton support (p_e), and with active exoskeleton assistance (a_e). The full demonstration is assessable in Supplementary Video S5.

Muscular Effort during Isolated Finger Flexion

To evaluate the impact of the exoskeleton on muscular effort, an experiment involving isolated index finger-, thumb flexion, and pinch grasp was conducted. The analysis utilizes an EMG-derived effort metric proportional to metabolic demand, computed as the integral of the squared RMS envelope, which has previously been demonstrated to correlate moderately with metabolic cost (Silder et al 2012). This experiment follows a modified version of the protocol shown in Figure 8**A**, while the motion and period order remain the same, the intervals were adjusted to a 1 s rest period followed by 5 s onset and 5 s offset periods. These longer onset and offset periods ensured that the motor actuator had sufficient time to fully bend the finger during the active exoskeleton condition, as a full range of motion requires approximately 2.5 motor rotations and lasts roughly 2 s. The evaluation was conducted on a single participant who performed ten repeated trials across the three operating conditions for both the index finger, thumb, and pinch while acquiring EMG to compute the effort metric. However, z-score normalization was replaced by maximum voluntary contraction (MVC) to scale the EMG demand relative to a scenario of pinch-lifting a 1 kg cylindrical object.

Muscular Effort during Object Grasping

The grasping experiment evaluated muscular demand during the grasping and lifting of a spray bottle under two conditions: without the exoskeleton and with active exoskeleton assistance. A new protocol was designed to capture this lifting behavior (see Figure 8**C**). The same participant performed ten repetitions of a grasping, lifting, placing, and releasing sequence in five second time intervals, which were coordinated with sound cues to initiate motion execution. To examine the effect of object weight, the spray bottle was filled with water to establish five loading conditions ranging from 0 kg to 1 kg in 0.25 kg increments, while the empty bottle weighed 278 g.

3.6. Demonstration of Real-Time Model Control of Pinch Capable Exoskeleton

This demonstration deployed the best-performing model and modality (the EMG CNN+LSTM architecture detailed in Section 4.3) for real-time assistive control of the exoskeleton. To assess the model's generalizability, the demonstration was conducted with a participant who was entirely unseen in the model training set. During the real-time session, the participant initiated a pinch motion, which was detected by the model to dynamically trigger the exoskeleton for assist grasping of everyday objects. The collection of everyday objects included a water bottle, pouch, sponge, screwdriver, and pencil. The full demonstration is assessable in Supplementary Video S6.

4. Results

This section presents the experimental results used to evaluate both the hardware and control components of the proposed system. First, the integrated exoskeleton sensors are characterized to examine the mechanical and electrical behavior of StretchSense and the tactile response of MagSense. The index-only exoskeleton configuration is then evaluated during grasping of everyday objects to assess how the integrated sensors respond during physical interaction. Next, the EEG- and EMG-based, among decision-level fusion classification results are presented for both subject-dependent and subject-independent conditions, followed by a real-time demonstration of hand motion classification. Finally, the assistive capability of the exoskeleton is evaluated through EMG-derived muscular effort during isolated finger flexion and loaded grasping, before demonstrating real-time model-based control of the index and thumb assisting exoskeleton. Statistical significance was evaluated using a Friedman test for repeated samples. This non-parametric test was used to compare three or more dependent groups, such as different models or experimental conditions, by ranking the data within each participant. When the Friedman test indicated a significant difference between groups, post-hoc comparisons were performed using the Wilcoxon signed-rank test. This non-parametric test compares two related paired samples to assess whether they are likely to originate from the same distribution.

4.1. Characterization of Integrated Exoskeleton Sensors

StretchSense

The characterization tests consisting of 31 cycles on the universal testing machine showed that all spring thicknesses exhibited a reduction in resistance when stretched following a pattern similar to what is is shown in Figure 9**A**. A video of the experiment is found in Supplementary Video S1. Figure 9**A** left shows the spring response during a 40 mm stretch-and-release cycle. As the spring is stretched, the generated force increases while the material resistance decreases, indicating an inverse relationship between force and resistance. A Wilcoxon signed-rank test was done on the 30 trials between the five different thickness springs. The Wilcoxon test revealed a statistical significant difference between their resistance and force across all thicknesses ($W \approx 0,\ p \ll 0.05$).

The 0.5 mm spring had the highest resistance, but also the widest sensing range between ~60 kΩ to ~40 kΩ, a range of 20 kΩ. It also had the lowest exerted force with a peak of ~0.7 N, whereas the 2.5 mm thick spring was the opposite. With the narrowest sensing range between ~6 kΩ to ~4 kΩ, a range of 2 kΩ, and the highest peak force of ~9 N. Of the five different springs, the 1 mm thick spring was chosen for final integration as it had a sensing range between ~20 kΩ and ~12 kΩ and exerted a peak force of ~3 N, offering a good compromise between restoring force and sensing range.

The 1 mm spring was further tested to characterize its transient behavior over 500 cycles. During the cyclic test, the spring failed after 455 cycles, the mean and standard deviation of the relative resistance change and force from these 455 cycles, is shown in Figure 9**A** left. Figure 9**A** right, shows the stress-strain response of the 1 mm spring during loading and unloading. At the beginning of the loading curve, the strain increases while the stress remains close to zero. This initial region can be described as a toe region, where slack in the spring is removed before the spring starts carrying significant load. After this region, the stress increases nonlinearly with strain, showing that the spring becomes stiffer as it is stretched. This strain-stiffening behavior indicates that larger forces are required to produce additional deformation at higher strains.

The unloading curve does not follow the same path as the loading curve, but instead lies below it. This difference between the loading and unloading paths shows that the spring exhibits hysteresis. The red area enclosed by the two curves corresponds to mechanical energy that is lost during one loading-unloading cycle, most likely due to internal friction or viscoelastic effects in the spring material. The hysteresis therefore shows that not all of the energy stored during stretching is recovered during relaxation. When looking at Figure 9**A**, the two plots together show, that the spring stays somewhat consistent throughout the 455 cycles, with the biggest deviation being in the beginning and the end of the stroke, where the material fatigue and memory would come to show. The plots also shows that the spring has ~40% reduction in resistance, between no stretch and full stretch, showcasing just how wide the range of sensing is for this particular StretchSense spring.

Actuator Response with StretchSense

Testing the actuator response together with StretchSense validates that the behavior seen on the universal testing machine is also observed when implemented and used on the exoskeleton. Figure 9**B** left, shows the behavior of StretchSense together with the motor position. Here it can be seen how the resistance of the sensor drops as the motor winds the tendon, contracting the index finger and thereby stretching the sensor. When the motor starts to unwind the tendon, the resistance increases again, but overshoots the baseline. It does however fall back to around its baseline towards the end of the relaxed period. Figure 9**B** right, shows the full motion, from fully relaxed to fully stretched, the result of which is showcased in Figure 9**B** left. A video of the experiment is found in Supplementary Video S1.

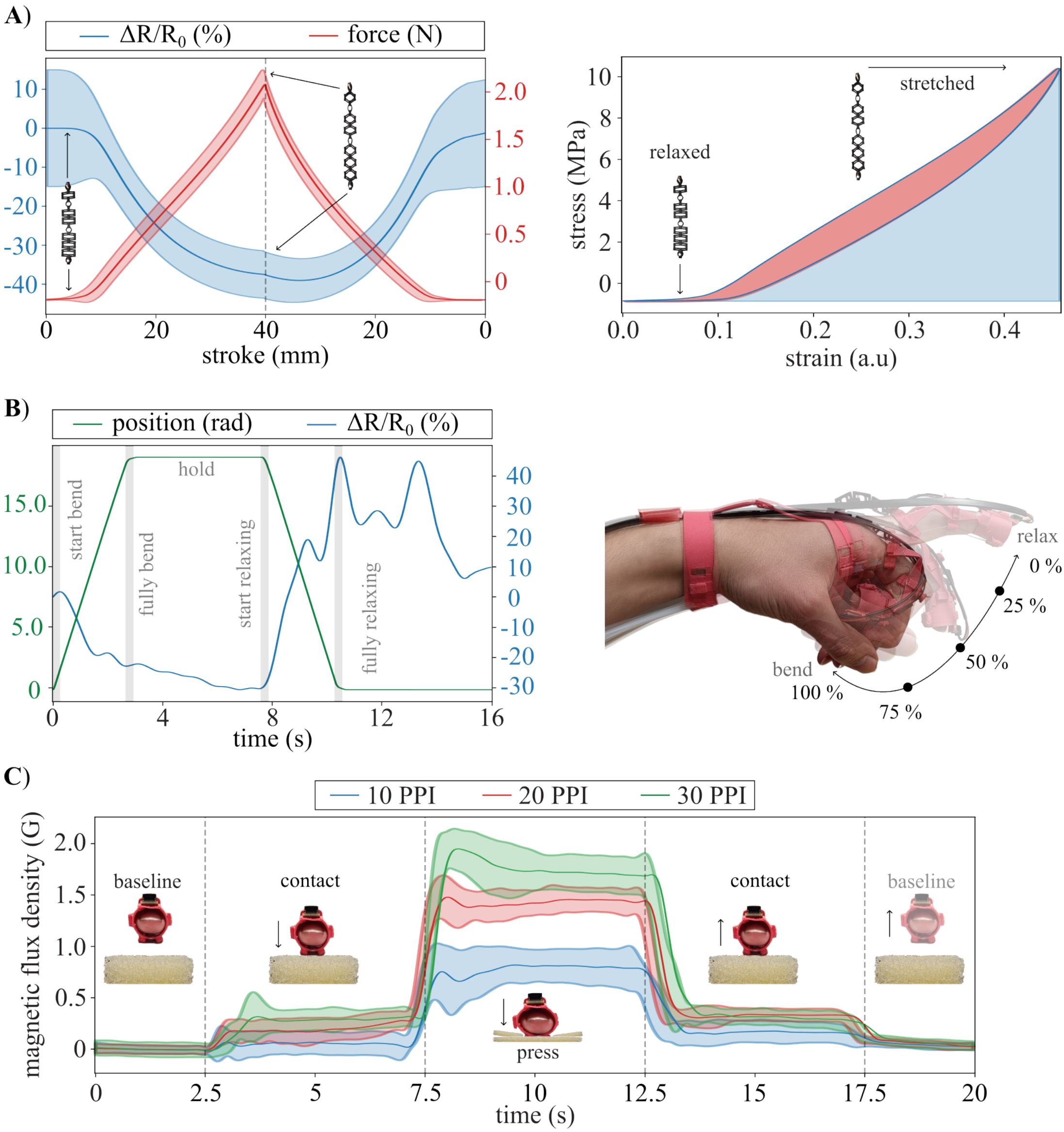


Figure 9. ***A*** *left, shows how the mean and standard deviation of the force and the relative resistance of the* 1 mm *spring changes when stretched to* 40 mm *and then relaxed back to* 0 mm*. The mean and standard deviation is calculated from the extended test of 455 cycles.* ***A*** *right show the resulting stretch-strain plot of the* 1 mm *spring.* ***B*** *left, shows how the motor position and the relative resistance change of StretchSense when used on the exoskeleton and how they correlate to each other during the motion shown on* ***B*** *right, where the motion from 0% to 100% bend is shown.* ***C*** *shows the results from the MagSense characterization test on the three variations of foam with 10, 20, and 30 PPI.*

MagSense

Testing MagSense on the three porosities of foam yielded the results shown on Figure 9**C** and the tests can be seen in Supplementary Video S1. On Figure 9**C** the filtered signals with mean and standard deviation, show the magnitude difference observed from baseline, to touched, to pressed and then back at baseline again. When MagSense is not touching the foam, the baseline is close to 0 G after the background noise has been accounted for. When the sensor is lightly placed on the foam, the signal increases to ~0.5 G indicating that there has been made contact with an object. Then, when the foam is compressed 10 mm, the signal increases to around 0.7 G, 1.5 G and 2 G, for the 10 PPI, 20 PPI and 30 PPI foam respectively, showing that there is a relationship between the firmness of the object and the magnitude of the signal from MagSense. This difference was also validated using a Wilcoxon signed-rank test on the section of data where the foam is pressed. The test revealed that there is a statistical significant difference between the signals produced by compressing the different foams with the same displacement ($W \approx 0,\ p < 0.05$).

4.2. Object Grasping using Index Module

Figure 10**A** shows the five different objects grasped using the index-only configuration. The objects were selected to represent different everyday geometries and mechanical properties, including rigid, slender, soft, and deformable objects. In this configuration, grasping was performed as a pinch-style grasp, where only the index finger was actively actuated against the object. The maximum values recorded during each grasp are shown for both the active and passive exoskeleton.

The results demonstrate that the index module was able to produce stable grasps across four of the five tested objects at a motor position of approximately 19 rad. Supplementary Video S2 shows the demonstration. The water bottle was the only case where actuation stopped earlier, at 16.8 rad. This was due to the water bottle being a large and rigid object, which produced sufficient fingertip contact to trigger the MagSense threshold before the motor reached the nominal end position. This result shows that the tactile feedback can be used to stop actuation when contact is detected, reducing unnecessary tendon tension after object contact. Figure 10**B** shows how the sensor responses vary depending on the object. The motor position follows the same general sequence for each trial, with the numbers 1 to 4 indicating motion onset, start of the hold phase, start of release, and return to rest, respectively. Although the actuation profile is similar across the the different objects, the sensor responses differ depending on the interaction between the fingertip and the object.

MagSense primarily reflects local fingertip contact and compression, while StretchSense reflects deformation of the actuated finger module during flexion. Together, these signals provide complementary information about the grasp: MagSense indicates contact with the object, whereas StretchSense gives information about the resulting finger posture and deformation. The screwdriver produced the largest StretchSense response, suggesting that this grasp required the largest deformation of the index finger among the tested objects. In contrast, softer or more compliant objects, such as the sponge, produced a different MagSense response because part of the applied motion was absorbed by deformation of the object itself. The variation in sensor response across the objects further demonstrates that the integrated sensing has potential for capturing differences in object interaction.

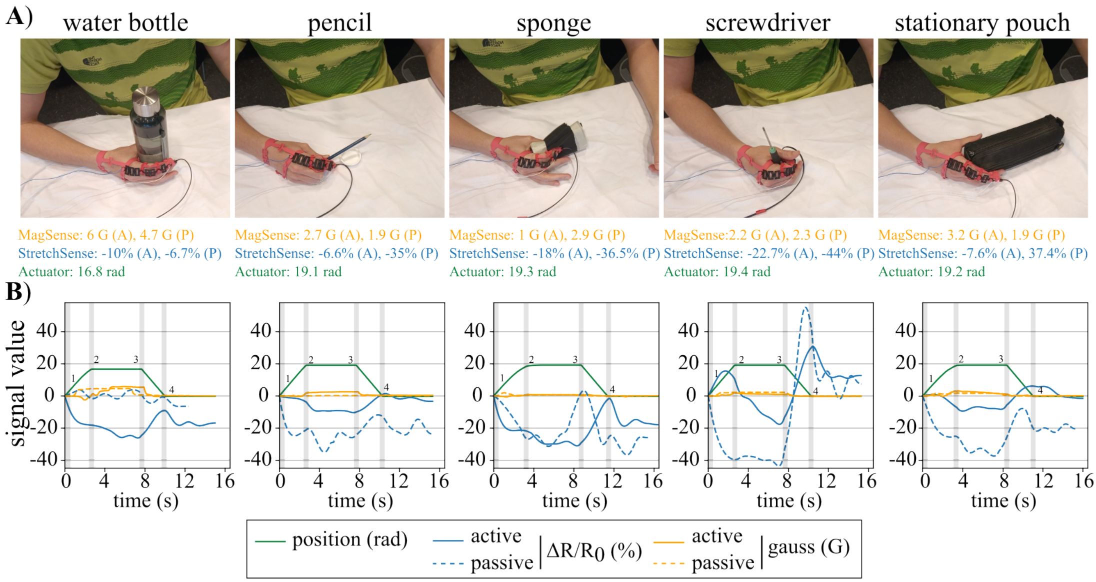


Figure 10. ***A*** *index finger exoskeleton grasping five different everyday objects, measuring the maximum magnetic flux, relative resistance, and position of the motor during grasping of each object.* ***B*** *shows the data acquired during grasping for the different objects. On the encoder position, the numbers 1 to 4 indicates the motion onset, start of the hold phase, start of release, and return to rest, respectively.*

4.3. Index and Thumb Flexion Classification

The following section shows the findings for subject-dependent and independent classification.
All results was visually inspected by histograms and Q-Q plots, revealing a violation of normality including outliers, leading to the statistical method chosen throughout this section to be the non-parametric Friedman test and Wilcoxon signed-rank test. The experimental protocol and setup can be watched in Supplementary Video S3.

Subject-Dependent Classification

The subject-dependent classification across networks using EEG is visualized in Figure 11**A**. It presents the test accuracy distribution for all 17 participants across each network. An inter-subject variability is noticeable within the same network, where N_1 achieved a mean and standard deviation accuracy of ($36.1 \pm 7.7\%$), N_2 ($47.2 \pm 14.1\%$), and N_3 ($56.7 \pm 10.0\%$). However, clamping the hierarchical network architecture offers significant advantages for the model to better distinguish between ME tasks. The Friedman test revealed significant difference ($\chi^2(2) = 27.6$, $p \ll 0.05$). Post-hoc Wilicoxon test showed that N_1 performed overall worse, compared to N_2 ($W = 8.0$, $p < 0.05$) and N_3 ($W \approx 0$, $p \ll 0.05$). Additionally, the test revealed a significant difference between N_2 and N_3 ($W = 1.0$, $p \ll 0.05$). The statistical test of $W \approx 0$ indicates a near-perfect paired comparison, where all subjects differs from the baseline network, N_1.

Subject-Independent Classification

The subject-independent classification, involves evaluating model performance to generalize across participants, and a comparison is conducted to examine which networks among modalities that contribute to the highest generalization. The statistical analysis is conducted on the validation accuracy. A distribution for the eight 8-fold cross validation accuracies and losses are visualized in Figure 11**B**. Observe how EEG has fewer correct classifications (Figure 11**B**, left) and less confidence (Figure 11**B**, right) compared to the other modalities. It is evident how the EEG modality suffers, lacking behind the EMG, likely due to inter-subject variability, low SNR, and less informative features. A significant difference is also found between the models ($\chi^2(2) = 13.0$, $p \ll 0.05$). The summarized representation given by LSTM alone is less effective compared to N_2 and N_3 ($W \approx 0$, $p < 0.05$). Whereas, for the remaining comparison, indicates that neither N_2 nor N_3 consistently outperformed the other across folds ($W = 11$, $p = 0.38$). The test accuracies in the order of N_1 to N_3 are approximately 34%, 48%, and 43%.

While comparing EEG and EMG accuracies (Figure 11**B**, left), a clear performance improvement is observed, with the mean accuracy increasing by approximately (41 – 50%). The EEG loss distribution are tightly clustered (0.94 – 1.08, with variability of ±0.01 – 0.15), whereas EMG loss distributions are substantially lower (0.20 – 0.33, with variability of ±0.26 – 0.37). This indicates how all models provide more confident when mapping corresponding EMG signals to tasks compared to EEG. The Friedman test revealed significance between models ($\chi^2(2) = 11.8$, $p \ll 0.05$). The N_1 lacks behind with a larger spread, where some folds perform worse than others, suggesting a more unstable model, that is less capable to generalize across participants, compared to N_2 ($W \approx 0$, $p < 0.05$) and N_3 ($W = 2.0$, $p = 0.02$). The spatial feature map obtained from the conventional neural network, handles inter-subject variability significantly better, which leads to improved regularity across folds with a reduced spread from the mean value, observe differences between N_1 and $N_2 - N_3$ in Figure 11**B** (left). However, allowing attention to search for relevance slightly lowers spread, but shows no significant performance gain ($W = 7.0$, $p = 0.38$). Regarding the LOSO inference test, reveals a ceil-limit performance for both N_2 and N_3, reaching a test accuracy of 99.4%, where N_1 lacks slightly behind with 97.8%.

In Figure 11**B** (right), the fusion loss mean and standard deviation fall within the intersection of the EEG and EMG distributions, showing that hyperparameter optimization successfully balances the minimization of both auxiliary modality losses. In Figure 11**B** (left), the accuracy on correct classifications tends to be unaffected by the uncertainty of EEG, leading to the decision-level fusion final classification to be mostly affected by EMG logits. The Friedman test showed significant difference for fusion across models ($\chi^2(2) = 13.0$, $p \ll 0.05$). Wilcoxon reflected similar trend as before, a significant performance gain from N_1 to N_2 and N_3 ($W \approx 0$, $p \ll 0.05$). While no significant between N_2 and N_3 ($W = 9.0$, $p = 0.25$). The test accuracies in the order of N_1 to N_3 are 98.9%, 98.9%, 99.1%.

Additionally, a Wilcoxon test was conducted to compare significance between modalities but within the same network architecture. The only non-significant difference found was between the EMG and fusion model when comparing across all models ($p > 0.05$).
All prior results revealed clear separation between modalities and performance gains depending on network architecture. EMG yields substantially higher validation and test accuracy than EEG across all models, approaching a performance ceiling. This limits the effectiveness of fusion, as the inclusion of EEG being less informative, offers little to no performance gain. This indicates that modality choice has a consistent impact on performance. Integrating the CNN architecture led to a more stable model with lower mean and less spread of losses. In contrast, the LSTM model exhibits both higher loss and greater variability. However, including the attention mechanism does not provide a clear benefit over the CNN+LSTM architecture in subject-independent classification.

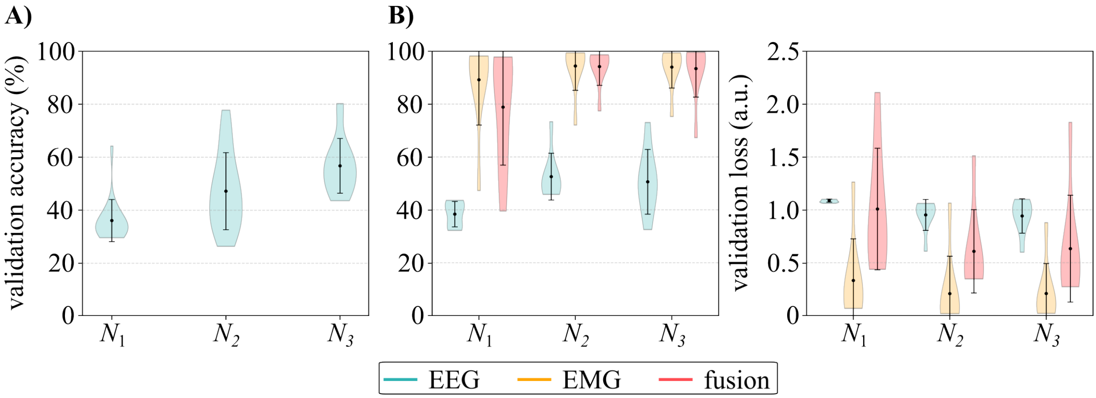


Figure 11. *This Figure summaries the index and thumb flexion classification results. The colors represent the modalities, where the violin separation presents the specific network, N_1 though N_3. **A** displays the subject-dependent classification results, where each violin distribution represents the validation accuracies across participants. **B** shows the subject-independent classification for validation accuracies (left) and validation losses (right). Each violin distribution represents the 8-fold cross-validation results.*

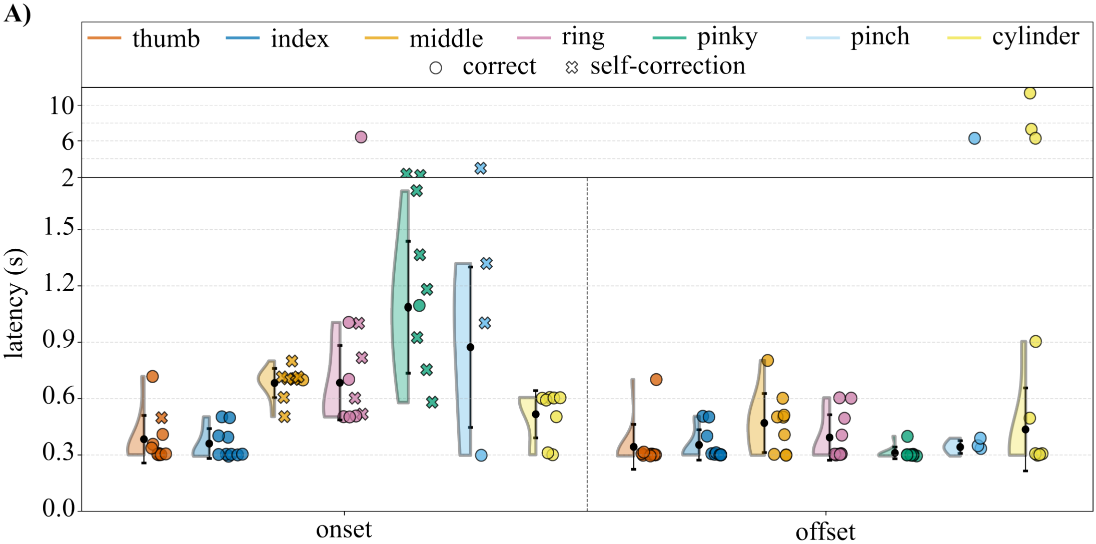
Based on these findings, the bare-LSTM model resulted in a lower mean accuracy and higher standard deviation across participants, whereas the added attention mechanism did not show a significant performance gain. The fusion of modalities happens to match the performance compared to the EMG unimodal, suggesting that informative EMG features are as reliable for classification. With that said, the CNN+LSTM architecture will be further investigated in the real-time classification system (Section 4.4) due to its high classification performance and simpler design. Furthermore, the system will be integrated solely with EMG acquisition to simplify experimental setup and time consumption.

4.4. *Demonstration of Real-Time Hand Motion Classification*

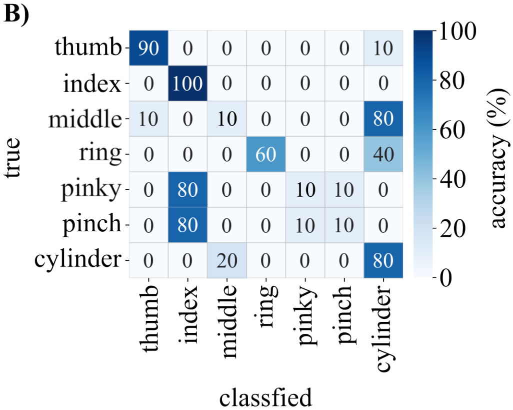
This experiment demonstrates the scalability of the EMG-based CNN+LSTM unimodal architecture in distinguishing all five individual finger motions alongside pinch and cylinder grasp motions during real-time operation. A digital twin designed in MuJoCo was utilized to visualize the model classifications in real time. The continuous, real-time behavior of the proposed system can be observed in Supplementary Video S4.

Figure 12 presents a breakdown of the model's performance, detailing classification latency (**A**), overall success rate via a confusion matrix (**B**), and the frequency of system self-corrections (**C**).
Illustrated in Figure 12**A**, is the evaluated system latency from when a participant's movement was detected (onset or offset) to the final classification.

A)



B)



C)

motions	number of self-corrections
thumb flexion	1
middle flexion	8
ring flexion	4
pinky flexion	8
pinch grasp	3

Figure 12. ***A*** *Latency distribution across onset and offset phases for each motion, distinguishing between immediate correct classifications (circles) and self-corrections (crosses).* ***B*** *Is the confusion matrix outlining classification accuracy across 10 trials for each motion. The vertical axis represents the ground-truth intent, while the horizontal axis denotes the model's classification.* ***C*** *present the total number of logged self-corrections events per motion type.*

While most motions achieved rapid execution within 0.3 to 0.7 s, specific fingers, such as pinky flexion and pinch grasp, exhibited higher onset latencies due to initial classification instability, which frequently required the system to self-correct (denoted by 'x' marker). Figure 12**B** illustrates the mapping of true vs. classified intents. A true positive was logged whenever the digital twin executed the exact motion performed by the participant. Conversely, if the model misclassified the intent, the corresponding incorrect motion was registered. As a general trend, index and thumb flexions demonstrated exceptionally high precision, matching the robust performance observed during the cylinder grasp, although two consecutive motions were misclassified in the process.

A recurring transient trend was observed during the trials: the digital twin occasionally initiated an incorrect motion but self-corrected to the intended motion. Figure 12**C** outlines the frequency of these self-correction events. A self-correction was defined as an initial misclassification that shifted to the correct intent, after which the subsequent motion remained in a stable, desired state. This behavior was particularly notable during middle and ring finger flexions, which were frequently mistaken for a cylinder grasp. It is suspected that by isolating these two fingers separately requires demanding physical execution while fixating the remaining fingers, leading to a high co-activation across multiple EMG channels temporarily resembling a cylinder grasp profile.

However, after a short duration, the signal transient settles, allowing the model to correctly recognize the isolated finger flexion. A similar transient event occurred during pinky flexion, where the initialization of the motion briefly resembled index flexion. However, a magnitude difference in the settled signal eventually allowed the system to self-correct to pinky flexion The pinch grasp exhibited the lowest overall classification accuracy. Due to its highly similar signal and channel activation profile to index flexion, the model misclassified the intent as index flexion 80% of the time, and spent 30% of that time self-correcting between index and pinky flexion. To mitigate transient misclassifications, the digital twin was designed as a finite-state machine, capable of overwriting active flexion/grasp motions during onset, it enforced strict rules during extension phases. The twin only responded to an extension command if it corresponded to the currently flexed motion. For instance, an index extension classification detected during an active pinch onset would be safely ignored, maintaining the twin's current physical state. This architecture ensured that only currently flexed motions could be extended. Under this paradigm, the extension motion failed to be recognized as the corresponding active motion three times during the cylinder grasp and once during the pinch grasp. However, all other extension motions yielded direct corrections. The model reached a 73.6% success rate, despite the 24 times the model self-corrected, with an average response time of 0.45 ± 0.79 s during onset, 0.77 ± 1.92 s for offset, and those occurrences where the model self-corrected 0.64 ± 0.70 s.

4.5. EMG-Based Evaluation of Hand Exoskeleton Assistance and Muscle Effort

This section presents the EMG-based evaluation of muscular effort under different exoskeleton operating conditions, these are referred as n_e: no exoskeleton, p_e: passive exoskeleton support, and a_e: active exoskeleton assistance. The analysis compares muscular effort during isolated finger flexion and grasping tasks to investigate how passive support and active assistance influence muscular demand. The experimental procedure for evaluating muscular effort can be watched in Supplementary Video S5.

Muscular Effort during Isolated Finger Flexion

Figure 13**A**, illustrates the average EMG signal across ten repetitions and channels, for isolated index finger flexion (left), thumb flexion (middle), while **A** (right) shows the corresponding unloaded pinch grasp. Figure insets present the muscular effort computed for each repetition. For index flexion, the p_e condition significantly increased muscular effort compared to n_e ($W = 8.0$, $p = 0.048$), while a_e reduced effort compared to both cases ($W = 1.0$, $p < 0.05$). Thumb flexion showed significant differences across all conditions ($W \approx 0$, $p < 0.05$). Specifically, comparing p_e to n_e revealed that index flexion with a passive exoskeleton slightly increased muscular demand due to the added resistance, whereas thumb flexion was reduced likely because the exoskeleton restricted the thumb motion, lowering recorded muscle activity. Nevertheless, a_e significantly reduced muscular demand in both cases. Conversely, the pinch grasp showed no significant differences between conditions ($\chi^2(2) = 1.3$, $p = 0.51$). Because unloaded pinch motion requires minimal effort, the signal merely fluctuated around the baseline noise floor, prompting an analysis of the loaded pinch grasp to evaluate exoskeleton assistance during more demanding tasks.

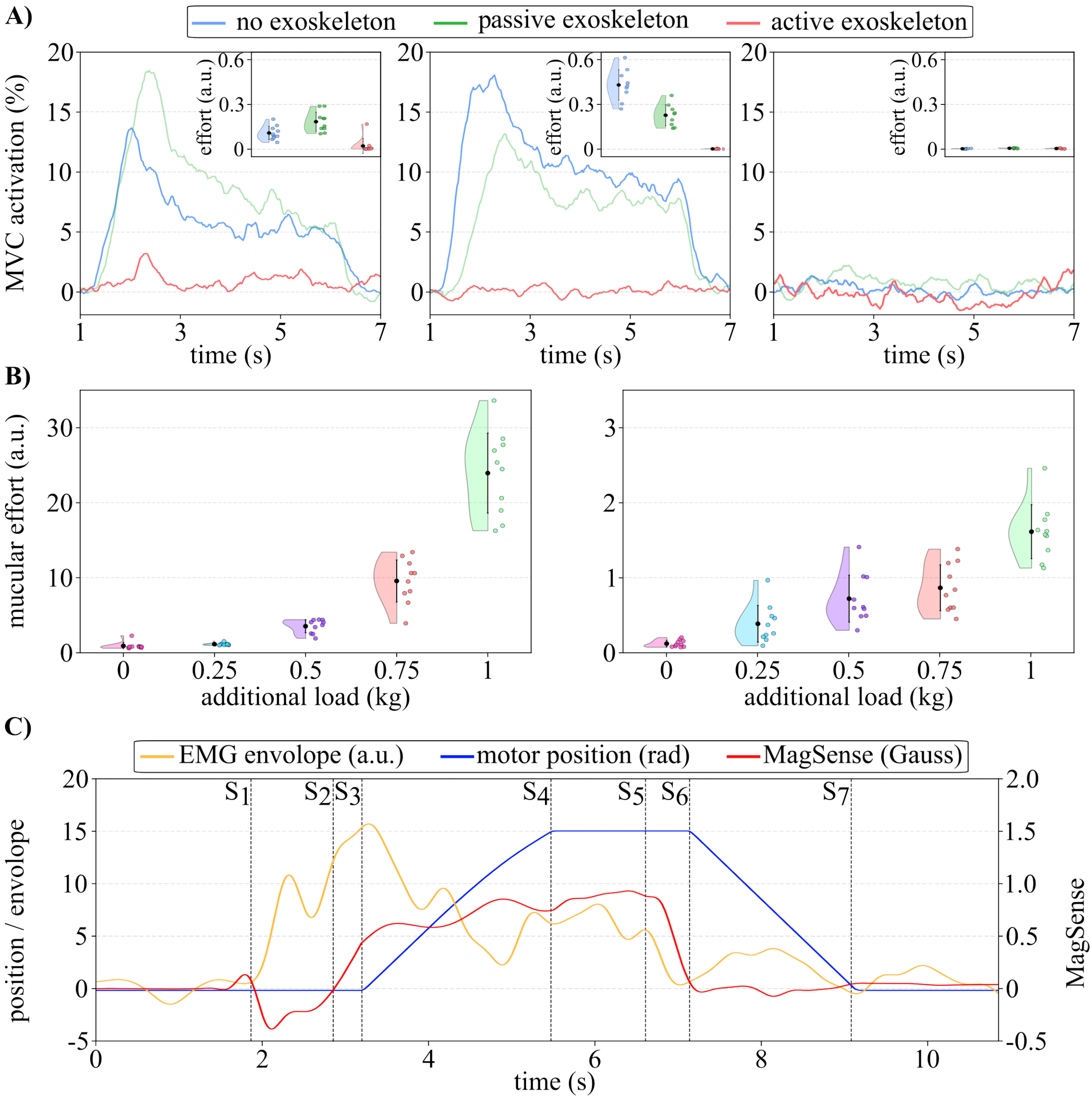


Figure 13. *This Figure summaries the exoskeleton assistance capability for isolated finger flexion and during object grasping, together with an overview of the system behavior across different states.* ***A*** *shows average EMG activity across trials and channels, normalized to the maximum voluntary contraction (MVC), for index finger flexion (left), thumb flexion (middle), and pinch grasp (right) under the three operation conditions indicated by the colors. The top right inset shows the corresponding muscular effort metrics, where each violin distribution represents individual flexion trials.* ***B*** *display the muscular effort for individual trials during loaded grasping of a spray bottle with added loads ranging from* 0 kg *to* 1 kg, *comparing the no exoskeleton (left) and active exoskeleton (right).* ***C*** *shows the system behavior during the real-time control of exoskeleton demonstration. The EMG envelope, motor position, and magnetic field response are indicated by colored signals across various system states,* S_1-S_7 .

Muscular Effort during Object Grasping

Figure 13**B** illustrates the EMG-derived muscular effort during repeated grasping and lifting of the spray bottle across increasing object loads for both n_e (left) and a_e conditions (right). For both conditions, muscular effort increased with additional loads, indicating a higher muscular demand during heavier lifting. Without the exoskeleton assistance, muscular effort increased exponentially from the unloaded object to the 1 kg added weight, where all except the two lightest conditions showed a significant difference ($p < 0.05$). On the contrary, a_e condition consistently reduced muscular effort across all loads. Even during the 1 kg lifting task, actively being assisted by the exoskeleton produced significantly lower muscular demand compared to the n_e 0.25 kg condition ($W \approx 0$, $p < 0.05$).

In general, the muscular effort was reduced by a factor of ten when assisted by the exoskeleton. The results further suggest that muscular effort under active assistance increased more gradually with additional load compared to the n_e condition. Where several neighboring loads between 0.25 kg and 0.75 kg, did not differ significantly, indicating that the exoskeleton partially compensates for the increasing lifting demand, but once exceeding the 1 kg the user is required to contribute with own force. Overall, the experiment demonstrated that active exoskeleton assistance reduced muscular effort during functional grasping and lifting across varying loads, with a reduction of 92.6% observed between the two conditions at the highest load.

4.6. Demonstration of Real-Time Model Control of Pinch Capable Exoskeleton

Supplementary Video S6 shows the demonstration of the model classifying motor intentions of an unseen person and controlling the exoskeleton, performing pinch grasps of different everyday objects. An extra demonstration of bilateral, human-to-human control were also performed, where one participant would use the biosignal classification pipeline to wirelessly control the exoskeleton worn by a second participant. This effectively allows an instructor or therapist to guide a participant's movement from across the room. The N_2 model was deployed in real-time to control the exoskeleton during active assistance. The participant was not included in the training data, showing that the model could work with an unseen user. During the demonstration, the participant initiated a pinch motion, which was detected by the model and used to activate the exoskeleton for grasping everyday objects. Figure. 13**C** illustrates the signal behavior of the EMG activity, motor position, and MagSense during grasping of a water bottle.

Initially, the participant remains in an idle state until S_1, where a voluntary pinch motion is initiated, causing an increase in EMG activity. At S_2, the MagSense detects contact with the object, while the continued increase in EMG activity enables the model to recognize the need for assistance. At S_3, the model classifies the motion intention and transmits a command to the ESP controller to actuate the motors. The motor position subsequently increases to approximately 15 rad and stops once the MagSense reaches a predefined threshold, indicating that the object has been fully grasped. The grasp is maintained until a voluntary release is initiated at S_5, resulting in a decrease in both EMG activity and magnetic field measurements. At S_6, the model recognizes the release pattern and assists the motors in releasing their tension. Finally, at S_7, the system returns to the resting state.

5. Discussion

This section interprets the main findings of the study in relation to the overall objective of developing a wearable biosignal-controlled hand exoskeleton for grasp assistance and evaluates how the mechanical design, integrated sensing, classification performance and latency, and real-time control contribute to the feasibility of the complete system.

Summary

In this study, we developed and validated SoFiE (Soft Finger Exoskeleton for Intelligent Grasping), a novel modular soft, underactuated tendon-driven hand exoskeleton designed to provide pinch grasp assistance. We also designed and evaluated a deep learning model to decode biosignals in real time, accurately identifying individual finger movements and grasp types. By closing the loop between deep learning biosignal processing, compliant tactile feedback sensors, and a lightweight physical actuator platform, the system demonstrates a translation of human intent into mechanical support.

The primary breakthroughs of this work can be summarized as follows:
The deep learning model successfully validated an online, real-time inference loop with a participant entirely unseen in the training phase of the model. Utilizing the N_2 (CNN+LSTM) architecture capable of recognizing finger flexion, extension, and prehensile movements at a high classification accuracy without recalibrating to specific users.

During physical evaluation, the exoskeleton significantly proved to offload muscular demand, while pinch-lifting a 1 kg loaded object, the exoskeleton yielded a 92.6% reduction in muscular effort (extracted by a EMG-derived effort metric). This drastically minimized the user's own energy expense, confirming the system's ability to assist pinch grasping.

Two novel sensors were developed and integrated as parts of the exoskeleton, these being the strain dependent resistive sensor StretchSense and the magnet-magnetometer tactile feedback sensor MagSense. StrethSense, while not actively used for control, did show that the change of resistance in the sensor could be used for detecting movement of the digit and possibly even used as a proxy for position estimation. MagSense proved to be able to distinguish between different firmness materials in a controlled setup opening the possibilities for tactile feedback to the user and the controller based on the object grasped. During runtime, MagSense was actively used to measure the magnetic flux, and if it reached a set threshold, due to a forceful grasp of a solid object, would stop actuation and thereby protect the user from discomfort.

Exoskeleton

The central design objective was that a compliant, modular, and underactuated exoskeleton could provide grasp assistance without any rigid joints and with single actuator. Since the hand is a complex mechanism (Taylor and Schwarz 1955), the design had to be general and adjustable enough to accommodate small differences between the design and the user, while still fitting tightly enough to transfer tendon forces effectively. The TPU glove structure with snap-fit modules supported this goal. The compliant material allowed the exoskeleton to conform to the user's hand, while the modular architecture made it possible to adjust and replace individual components without redesigning the entire device. The integrated spring structures further allowed local stretching during wrist and finger motion, helping the device remain low-profile and less restrictive on the hand.

The chosen actuation mechanism was thin steel tendons, which were routed in a similar way to the natural flexor tendons. This simplified the mechanical design as it did not require precisely aligned rigid links or revolute joints. Instead, the tendon tension, compliant glove structure, and natural kinematics of the hand together produced assisted flexion motion. By using thin 3 ply steel wire as tendons, it was possible to route the actuation path along the fingers and across the palm, while keeping the hand mounted structure compact. However, the routing path had to be kept simple to reduce the friction the wires would otherwise create, especially since the actuator was placed remotely from the hand. The tendons were capable of transmitting the force produced by the motor, but a fundamental limitation of this actuation method is that the tendons can only transmit a tensile force. Because of this limitation, only the flexion could be actively assisted, whereas the extension had to rely on the natural restoring forces of the user's finger and the StretchSense spring.

By having only a single actuator in the design, it supported the goal of keeping the system simple, lightweight, and wearable. For the index only version, the actuation was simple, pull the tendon with the actuator and the finger flexes. When adding the the thumb module, added friction in the tendon routing and a more difficult control problem was revealed. But, by using a floating pulley as a passive differential, the same motor could tension both the index and thumb tendon branches and produce a coordinated pinch motion. This kept the system simple, even with an added finger, but also came with its own trade-off. Because both tendon branches are tensioned by the same actuator, the index finger and thumb cannot be controlled independently. The passive differential mechanism can redistribute the motion depending on the resistance encountered by each digit, which allows for passive adaptation to object geometries, but does not provide individual digit control. For pinch grasp assistance this is fine, but tasks requiring independent or highly dexterous thumb and index motion, a different mechanism is needed.

To complement the underactuated mechanical architecture without adding excess weight and bulk to the user's hand, the two StretchSense sensors were integrated directly into the structural design of SoFiE, as described in Section 2.2 (StretchSense). Tactile feedback from the interaction point between the index fingertip and the object is provided by MagSense, which utilizes an embedded neodymium magnet and a magnetometer to measure the deformation of the fingertip during contact (see Section 2.3, MagSense and Figure 9). Through testing, it was shown that MagSense could distinguish between different porosities of foam (as shown in Figure 9**C**), in a controlled scenario, indicating that the system is capable of moving beyond contact detection and register distinct magnetic flux gradients depending on the object compliance. Outside of the foam test, the results still follows the behavior seen in the characterization, though with more uncertainty.

The readings from MagSense, depends on the firmness of the object as established before, but also on the quality of the grasp. If the magnet in the index module is not flat against the surface or there is other small variations in the grasp, it can be seen in the data. Also the force of which the user or the exoskeleton tries to grasp has a direct effect on the magnetic flux density due to the difference in displacement of the tissue, which is exactly what the sensor should be able to measure. The data from the demonstration in Figure 10**B** shows this variation, but also that the trends are similar, proving that with some tuning and more advanced logic, possibly a slight adjustment of the design of MagSense, there is real potential for high precision tactile feedback. Lastly the magnetic field that make the sensor work, is also what limited the MagSense to be used on only one of the two fingers, as they would otherwise interfere with each other during pinch grasps.

Proprioceptive feedback is at the same time handled by StretchSense on the dorsal side of the fingers. StretchSense is fabricated from conductive flexible TPU and serves a dual-purpose role: mechanically, it acts as an extension spring that stores elastic energy during flexion to passively return the digit to its neutral state. Electrically, it behaves as a strain-dependent resistor. As demonstrated in the characterization results (Figure 9**A**), the material exhibits a viscoelastic hysteresis loop that prevents it from immediately returning to its baseline during rapid unloading cycles. However, it displays a predictable and repeatable ~40% reduction in resistance from rest to full stretch, mapping reliably to the physical flexion state of the exoskeleton.

Although StretchSense was not integrated into the active control loop for this iteration, its performance confirms its viability for continuous, "sensorless" position estimation in future design iterations. This is also backed by the results shown in Figure 10**B**, where the behavior of StretchSense followed the degree of motion required for grasping the five different objects. There were a difference between the active and passive results. It is possible that this is due to the spring not being under a slight pretension in the current design, meaning the resolution is reduced and it is mostly working in the bottom part of the stretch strain curve (see Figure 9**A** right, where the toe region is). Comparing the two plots in Figure 9**A** the toe region lies within the same area as where the resistance of the sensor has the biggest standard deviation, resulting in bigger variations with small grasp than with larger movements. A combined table of specifications for the exoskeleton is shown in Table 4.

Table 4. *SoFiE specifications.*

Attribute	**Value**
Weight	Total: 354 g Glove: 14 g Enclosure: 340 g
Power	Total power draw: 1.7 W ESP32: 1.2 W Actuator: 0.48 W Buck converter efficiency: 85% Battery power: 7.5 Wh
Battery life	$\frac{7.5Wh \cdot 85\%}{1.7W} = 3.75$ hours
Sensor range	MagSense: -8 to +8 G StretchSense: 12 to 20 kΩ

Network Architecture, Modality Performance and Fusion

In the early stages of this study, it was hypothesized that multimodal fusion, the integration of discriminant information acquired from EEG and EMG, would complement the weaknesses of each modality and provide more robust, invariant features to improve model generalization across users. EEG signals allow direct access to cognitive perception, including motor planning and execution, offering the potential of remotely controlling devices through thoughts alone. However, despite the intriguing concept of mind-to-action control, EEG struggles to achieve stable, invariant, and discriminant information. This is because the non-invasive nature of capturing bioelectric potentials through the cranium, tissues, and skin causes a low signal-to-noise ratio (SNR). Additionally, intra-subject variations, driven by the highly dynamic nature of brainwaves and their constant shifts based on psychological and neurophysiological factors over time, create a severe disadvantage for BCI systems seeking to accurately understand how these changes are linked (Huang et al 2023).

The evaluation of the electroencephalogram (EEG) modality was investigated across 17 participants, highlighting the discriminant information this modality provides across the proposed networks ($N_1 - N_3$) for the ideal scenario of designing a tailor-made model for specific users (Section 4.3, Subject-Dependent Classification). Previously, this approach was shown to have less time-frequency response variation, leading to better model performance (Huang et al 2023). Across the participants, the EEG modality achieved a low-to-moderate success rate under the full deployment of all integrated networks (N_3), resulting in a 56.7 ± 10.0% classification accuracy (see Figure 11**A**). This highlights how individual neurophysiological responses differ, with some users being more capable of producing consistent and detectable sensorimotor rhythms (Huang et al 2023; DH Kim et al 2023).
It was later observed that allowing the model (with the corresponding N_3 architecture) to learn discriminative features across all participants decreased the validation accuracy to 50.60 ± 11.46% under 8-fold cross-validation (see Figure 11**B**) due to inter-subject variability.

The subject-dependent evaluation additionally proved that utilizing the full hierarchical network architecture significantly benefited the model in distinguishing between index and thumb motions, yielding the highest peak user performance of 80.25% and the largest average performance gain across all participants. It is suspected that the attention mechanism more efficiently captures rhythmic behavior and learns to concentrate on relevant temporal features, which more effectively explains the ongoing motion and discrimination between classes. However, during the subject-independent evaluation, the N_3 architecture was unable to deliver superior performance, leading to a non-significant difference between N_2 and N_3 ($W = 11, p = 0.38$) across the 8-fold cross validations. This indicates that regardless of the proposed architectural optimizations, the highly inter-subject feature variance establishes a performance ceiling. This interpretation is corroborated by the EEG spectral analysis (see Figure S1**A**), which demonstrates that signal intensities across cortical regions and frequency bands differs drastically across participants.

The EEG spectral analysis lead to the channel selection of the central (C3, Cz, C4) and frontal (F7, F3, Fz, F4, F8) regions with the frequency range of 0.5 – 30 Hz. The combination of these achieved the highest class separation properties, although a weak separation was observed between flexion and extension movements (See Figure S1**B**, onset vs offset). It is suspected that these motions are co-localized within the same small region, and given the poor spatial resolution of EEG, the motion signals appear highly similar. The topographical visualization in Figure 4**E** confirms similar activation patterns in the central region during both onset and offset. Previous state-of-the-art findings highlight similar occurrences and underscore the challenge of distinguishing distinct movements within the same limb (Yong and Menon 2015).

In contrast to the cortical limitations observed with the EEG modality, electromyography (EMG) provides a direct measurement of motor execution by capturing the electrical activity of muscle contraction. Because EMG is directly mounted on the surface of the skin, it is less affected by attenuation compared to EEG, it inherently yields a significant higher SNR and it offers more robust discriminative features for the motion recognition model. The evaluation across the proposed networks demonstrated the strong potential of EMG for motion classification as displayed in Figure 11**B** (left). It highlights how the spatial-temporal feature extraction provided by the N_2 architecture was already a prominent decoding method, achieving a validation accuracy of $94.4 \pm 8.62\%$ across 8-fold cross validations. However, the optimized version (N_3) delivered identical performance, yielding a non-significant difference between N_2 and N_3, both achieving an exceptional performance on the leave-one-subject-out (LOSO) test, getting a 99.4% test accuracy. Standalone EMG features establishes an early and near-perfect performance ceiling, even for the baseline LSTM model but slightly lacking behind (97.8% test accuracy), even though the spatial-temporal features have proved to be sufficient for motion distinguishing tasks and less computational demanding compared to stacking the attention mechanism.

It was initially hypothesized that the multimodal fusion evaluation would examine how the cognitive planning of EEG alongside the direct muscular activation of EMG would form a more resilient model, but the results disproved this synergy. The integration of modalities into the decision-level fusion framework, showed similar non-significant difference to the standalone EMG ($p > 0.05$), while EEG was significant compared to both EMG and fusion for all networks. This indicates that the model remain unaffected by the uncertainty of the EEG signals, while solely relying on EMG representations, given the EMG validation losses offer lower error between classified signals and the ground-truth compared to EEG (see Figure 11**B**, right). It is also expected that the EMG performance ceiling limits the effectiveness of fusion frameworks, and in an alternative scenario where EEG offers +90% accuracy, the finalized multimodal would remain insignificant.
Despite the EEG unimodal and multimodal results, the preprocessing techniques and spatial-temporal features from N_2 showed a promising approach of decoding index and thumb motion, assesses a high-performance platform using muscular activation. Based on these statements, the EMG CNN+LSTM unimodal yields an optimal trade-off, providing less computational complexity and equivalent success rate compared to adding the attention, and requires less preparation time compared to including the EEG. With that said, this model was further investigated for scalable and real-time inference, among real-time exoskeleton control.

We initially hypothesized that introducing the architectural complexity of a CNN and an attention mechanism on top of the LSTM backbone would improve model robustness and generalization. This hypothesis proved partially correct. For the EEG subject-dependent evaluation, the attention mechanism was crucial, yielding a significant performance gain over the other two configurations and showcasing the network's ability to concentrate on relevant waveform rhythms related to motion execution. Conversely, when extending the framework to subject-dependent evaluation, the CNN+LSTM architecture proved just as efficient as the attention-augmented model. Essentially, across all evaluation paradigms, the standalone LSTM network was unable to match this performance. Which underscore that while a base LSTM is less sufficient for complex biosignal decoding, integrating convolutional features and attention mechanisms provides the necessary structural bias to navigate both subject-specific features and broader generalized motor patterns.

The unimodal EMG CNN+LSTM model was evaluated for scalability and real-time inference to examine its capability to distinguish between the flexion and extension of all five fingers, as well as the contraction and release motions of a pinch and cylinder grasp. The evaluation results were derived from observations during the demonstration, which can be viewed in Supplementary Video S4. Based on 70 performed repetitions, the model archives a success rate of 73.6% and response time of 0.45 ± 0.79 s on successful motion recognitions. Comparing this performance with the corresponding success rate during inference testing (i.e., evaluating the model on unseen data following the training phase) revealed a significantly higher test accuracy of 97.2%. We suspect that slight variations in EMG sensor positioning, combined with the fact that the model was trained on a single participant, caused the architecture to overfit to highly specific motor execution patterns.

Within a controlled environment, executing 90 continuous trials sequentially allows a participant to become remarkably precise, resulting in invariant movements. Incorporating a diverse cohort of participants into the training pipeline is expected to improve the model's generalization capabilities across subtle changes in sensor placement, applied force, and individual motor execution. During the real-time inference is was observed how index, thumb and cylinder motions exhibit rich and discriminant signal profiles. This distinction allows the model to easily and confidently classify these movements within the earliest stage of response, achieving a latency of 300 ms. Although the cylinder grasp remains a dominant class, the initial transition during the grasp generates signal profiles similar to those of middle and ring finger flexions. In the worst cases, rapid fluctuations in the model's output require a larger temporal buffer to establish a stable majority vote, thereby introducing a latency penalty of approximately 300 ms (compare the distributions between cylinder and index/thumb in Figure 12**A**). This similarity explains why middle and ring flexion are initially misinterpreted as cylinder grasps, where the model self-corrects shortly afterward once the signal settles.

A similar occurrence is observed between the pinky-, index finger, and pinch grasp, with the index finger generally remaining the dominant class. Implementing a priority scaling mechanism within the majority voting scheme proved highly effective at ensuring rapid, stable transitions when returning the motion to its initial resting position. The average response time for motion deactivation was 0.77 s. However, the standard deviation reached 1.92 s, a variability primarily driven by four extreme outliers. It was observed that these outliers were susceptible to being misinterpreted as middle finger extensions. This was likely due to EMG crosstalk and a prolonged signal discharge profile that resembled an middle finger offset movement.

Latency and Controller Logic

The findings of this work demonstrate a real-time exoskeleton control framework utilizing EMG signals. When deployed for real-time inference, the scaled model achieved a response time of 0.45 ± 0.79 s. This latency profile ensures a more stable output by requiring a majority voting buffer and a classification confidence threshold above 90%. However, reducing this latency remains a primary focus for future work to accelerate response times, which is essential for high-speed, fine-motor exoskeleton finger control. Conversely, for general upper-limb assistance, timing constraints tend to be more flexible depending on the specific hardware platform. The SoFiE platform, for instance, is less sensitive to minor latency delays when acting as an assistive grasping device. Instead, its primary value lies in its flexible, comfortable design and its ability to significantly reduce user muscular effort. Furthermore, because SoFiE utilizes an underactuated mechanical system, the classifier's complexity can be simplified to distinguish only between basic flexion and release motions.

The system is split into a client/server paradigm as shown in Figure 6, which enables development of the high-level and low-level controller separately. While the system as a whole, successfully decodes user intent and translates it into a physical assistance, the low level controller relies on a straightforward, bang-bang control approach. It uses the motor encoder position to only run within software set limits and treats the MagSense on the index finger as a binary contact switch to stop motor actuation upon reaching a set threshold (Section 2.7, Control Architecture). The simplistic controller did prove to be robust enough to deliver proof-of-concept validation, yielding a 92.6% reduction in user muscular effort during pinch-lifting tasks (Section 4.5, Muscular Effort During Object Grasping). But the current bang-bang strategy relies on abrupt changes in motor direction which can feel unnatural for a human-in-the-loop system, and it has no real way of dynamically adjusting itself during assistive motions.

Given that SoFiE features a fully integrated array of both tactile sensors (MagSense) and proprioceptive flexible sensors (StretchSense), the underlying hardware platform is inherently capable of supporting more advanced low-level control strategies. Instead of restricting the sensors to only be used for post analysis or simple safety cutoffs, they could be included in the real-time control loop. Following the review done by Proietti et al, more advanced controllers that SoFiE could transisiton towards fall into two categories: partially assistive controllers and standard passive and position feedback controllers. In the category partially assistive controllers, using admittance control or assistance-as-needed seems like the best options, as they are highlighted as good options for exoskeletons that lack back-drivability and are already commonly used control algorithms. PD and PID controllers are also commonly used to track trajectories dynamically calculated from a higher level controller or preplanned during offline training, and would also make a great option for controlling SoFiE (Proietti et al 2016).

The low-level controller could scale the motor velocity and tendon tension dynamically, by adjusting the assistance based on the EMG signals (Lenzi et al 2012), or modulate it based on the physical resistance encountered by the fingers during flexion. With MagSense and StretchSense alongside a more advanced controller there is real potential for low-level multi-sensor data fusion by continuously cross-referencing the profile of StretchSense with the signal from MagSense. That way, the controller would be able to actively estimate object compliance in real-time and adjust its assistance accordingly. Fusing the sensors would allow the exoskeleton to act in a closed-loop system, possibly even allowing for slip detection and autonomous grip adjustment and the floating pulley would still be an integral part of the actuation. It would assist the underactuated system in passively conforming to the geometry and ensuring a stable grip, that the system can then tighten or loosen depending on the sensory information.

Clinical and Practical Implications

Although the presented work was not evaluated on population with RA, the muscular effort results indicate a promising practical application for future assistive hand exoskeletons. The most relevant finding is that the active assistance significantly reduced the muscular effort required during the loaded grasping tasks. During the highest load condition, the active exoskeleton reduced the muscular effort by 92.6% compared to the no-exoskeleton condition (Section 4.5). This suggests that SoFiE can offload part of the muscular demand required to perform and maintain a functional pinch grasp, which could be relevant for users with reduced grip strength, joint pain, fatigue, or otherwise impaired hand function. For these users, the goal is not necessarily to directly increase their grip force, but to reduce their effort required to perform repeated or sustained grasping tasks during activities of daily living, and in turn increase their overall grip strength over a longer period of time.

The isolated finger flexion tests also highlight an important difference between the passive support and the active assistance (Figure 13**A**). During index finger flexion, the passive exoskeleton condition increased the muscular effort compared to the no-exoskeleton condition. This indicates that the exoskeleton introduces a slight resistance when it is not active, which is expected with any wearable device, especially when it adds tendons and springs to the hand. In contrast, the active exoskeleton condition reduced the muscular effort, showing that the actuator was able to overcome the added resistance and add meaningful assistance.
For thumb flexion, the passive exoskeleton condition reduced the muscular effort, which we hypothesize is because the exoskeleton restrict the motion of the thumb, not assisting it, emphasizing that the reduction in EMG activity and muscular effort must be interpreted together with the physical behavior of the device. For pinch grasping no significant difference between the three conditions was observed, when doing isolated movements, indicating the movements are too small to clearly demonstrate the difference between the conditions.

Since the pinch motion itself had an activation close to the noise floor of the EMG sensors, the room for the exoskeleton to improve the effort further remained limited. The loaded grasping experiment therefore provides a better evaluation of the assistance, since the task required the user to generate and maintain grip force while lifting the object. When lifting the loaded object, the muscular effort increased with the load in both conditions, but the increase was more pronounced without the exoskeleton compared to with the active exoskeleton. With the active assistance, the effort remained substantially lower across all tested loads and increased more gradually, indicating that the exoskeleton compensated for most of the increasing load (Figure 13**B**).
From a practical perspective, the design of the exoskeleton is also important. By moving the heavy electronics and actuation parts to an upper-arm enclosure, the weight mounted on the hand is reduced to only that of the glove structure, tendons, and integrated sensors. For assistive devices, where additional weight on the hand or wrist could increase fatigue, restrict motion, or in general reduce the comfort, this is an important factor. The lightweight glove therefore supports the overall goal of creating a wearable system that can provide assistance without obstructing the natural hand movements in any meaningful way.

6. Conclusion

In this work, the development, hardware characterization, and validation of the soft wearable hand exoskeleton, SoFiE, have been successfully demonstrated. The mechanical implementation proves that utilizing variable-shore 3D-printed TPU combined with an underactuated, tendon-driven floating pulley mechanism successfully addresses the challenges of bulky, rigid hand orthoses. By eliminating the necessity for strict biological joint alignment, the device ensures a highly compliant and comfortable fit suitable for daily assistive tasks. Furthermore, the successful integration of the dual-purpose StretchSense component and the MagSense fingertip tactile sensor shows that proprioceptive feedback and contact-aware safety functions can be embedded directly into flexible structures without adding rigid components or limiting user mobility.

On the control side, comprehensive offline and online evaluations provided vital insights into the interplay between architecture complexity and biosignal choice. Although we hypothesized that an EEG-EMG multi-modal fusion network would achieve superior classification accuracies, experimental evidence revealed that low neural SNR and severe inter-subject variability cause the decision-level fusion model to achieve no significant statistical advantage over an EMG unimodal configuration. However, our hypothesis regarding architectural complexity was fully validated: integrating a 1D CNN for spatial-temporal feature extraction prior to sequential LSTM processing significantly stabilized the training curvature, lowered classification losses, and generalize more effectively across participants compared to standalone recurrent networks. When deployed under strict real-time constraints, the resulting EMG-based CNN+LSTM pipeline scaled successfully to a complex 15-class configuration, highlighting the potential of fine-motor recognition control.

The cooperation between SoFiE and EMG-based CNN+LSTM model inference, driving assistive commands smoothly via a TCP client/server network with low operational latencies. Ultimately, human-in-the-loop muscular evaluations confirmed that SoFiE meets its primary objective. While wearing the device passively preserves natural hand mechanics without introducing excessive load, activating the exoskeleton delivers highly effective mechanical assistance, resulting in a remarkable 92.6% drop in muscle effort during demanding lifting tasks. Collectively, these results demonstrate that combining lightweight, sensorized soft robotics with robust deep-learning intent-decoding frameworks offers a highly viable, scalable pathway toward restoring fine-motor autonomy.

Future Work

Future work for SoFiE should focus on improving the physical glove design to make it easier to take on and off, particularly for users with impaired hand movement. The modular architecture would also allow for expansion to additional fingers, enabling assistance beyond pinch grasping and supporting a wider range of hand gestures.

Both MagSense and StretchSense showed promising sensing behavior, indicating strong potential for integration into the real-time control loop. With reduced slack in the StretchSense structure, the sensor could provide more reliable estimates of finger posture during actuation. MagSense could be used as the primary tactile feedback unit for contact detection, grip-force modulation, and contact-aware safety. Combined with a more advanced low-level controller, such as an assistance-as-needed controller, these sensing modalities could allow the exoskeleton to adapt its assistance based on both user intention and object interaction. These improvements would support closer integration with the deep learning-based control framework and move SoFiE toward a full-hand assistive device for gesture-based grasp assistance.

On the model inference side, will focus on improving the speed and performance of the real-time model. Specifically, we plan to experiment with different sliding window and buffer step sizes. This will show us how changing the data update frequency affects the model's ability to quickly catch new muscle activations and how our different networks handle these changes.
Additionally, we will test more network combinations to find the best setup. We want to evaluate alternative designs combinations, such as a standalone CNN, base-attention mechanism, LSTM+attention, and CNN+attention. We will benchmark these architectures using individual modalities along decision-level fusion.

To fix the long delays (outliers) seen during grasp releases, we plan to add another EMG sensor on the back of the forearm. This sensor will capture the quick muscle burst that happens when a user opens their hand, helping the model realize the movement has ended. Finally, we will test the real-time system with more participants to ensure the model can handle slight changes in sensor placement and different muscle strength profiles.

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